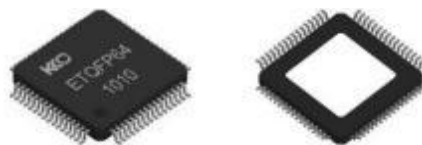


Features

- AEC-Q100 qualified. (on-going)
- Normal operating battery supply voltage 5.5 V to 19 V
- Normal operating VDD supply voltage 4.75 to 5.5 V
- Temperature range -40°C -135°C
- 10 integrated PowerMOS are controlled SPI or INx:
 - 4 high/low side PWM with $R_{ON} (max) = 0.3\Omega @ T_j = 150^\circ C$
 - 6 low side PWM with $R_{ON} (max) = 0.3\Omega @ T_j = 150^\circ C$
 - 1~20kHz PWM frequency range
- 4 integrated PowerMOS are controlled SPI :
 - 4 low side PWM with $R_{ON} (max) = 0.3\Omega @ T_j = 150^\circ C$
 - 1~20kHz PWM frequency range
- Current sense amplifiers
- Enable Driver (EN_DR) to used to enable or disable the operation of all channels
- RESn(Reset) used to disable all channels and resets internal registers to default values
- Logic inputs TTL/CMOS-compatible
- Output voltage clamping 37 V typ. in low side
- SPI interface for outputs control and for diagnosis data communication
- Over temperature protection
- Open load, short to GND, short to VB
- Overcurrent diagnostics in latched or unlatched mode for each channel
- Controlled SR for improved EMC behavior
- Full ISO26262 compliant, ASIL-D ready

Description



The HE9314 is a flexible(Serial Peripheral Interface+ PWM) controlled 14 channels of configurable, monolithic solenoid driver IC including 4 high/low controlled half bridges by Inx or spi, 6 low side controlled by Inx or spi, and 4 low side controlled by spi, all with the possibility of PWM control.

Current sense amplifiers are integrated for sensing voltage drops across current-sense resistors at common-mode voltages from 5.5 V to 40 V, independent of the supply voltage.

HE9314 can execute closed-loop control by MCU ADC and PWM resources, targeting the solenoid control for Integrated Hydraulic Braking, Electronic Stability Control, Active Suspension, etc.

Through the SPI it is possible to configure the device parameters like configuration, Slew-rate, Overcurrent threshold, to send the drivers commands and to read back the diagnosis results.

Application

- ESC
- EHB
- ASU



Part Number	Package	Marking	Packing
HE9314RIAD	ETQFP	HE9314	1000/tape&reel

- *Not sure if you got the latest datasheet or not. Please kindly contact the sales of KKChips to confirm. Thanks.

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1 Block diagram and pin description

1.1 Block diagram

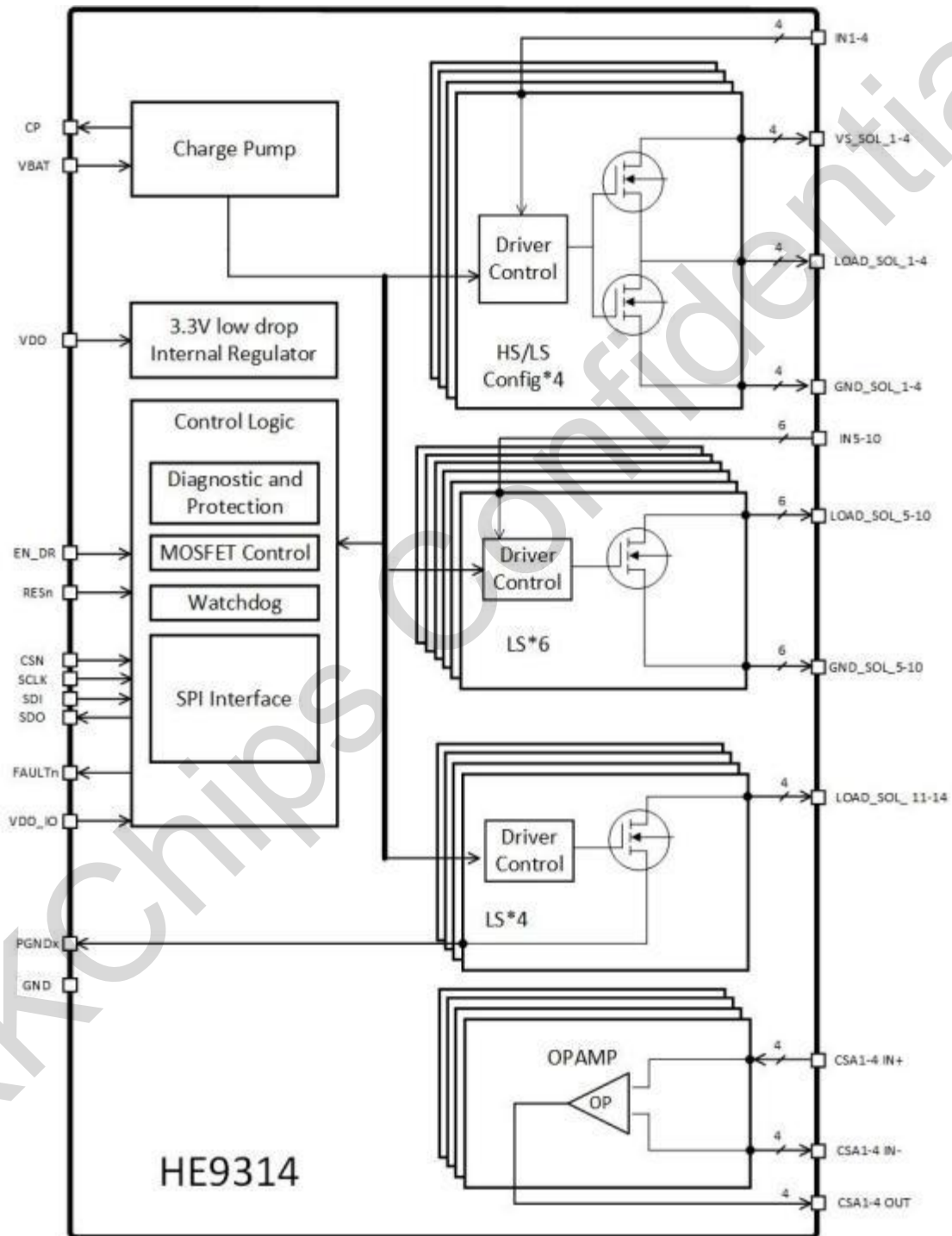


Figure 1-1 Block diagram

1.2 Application schematic

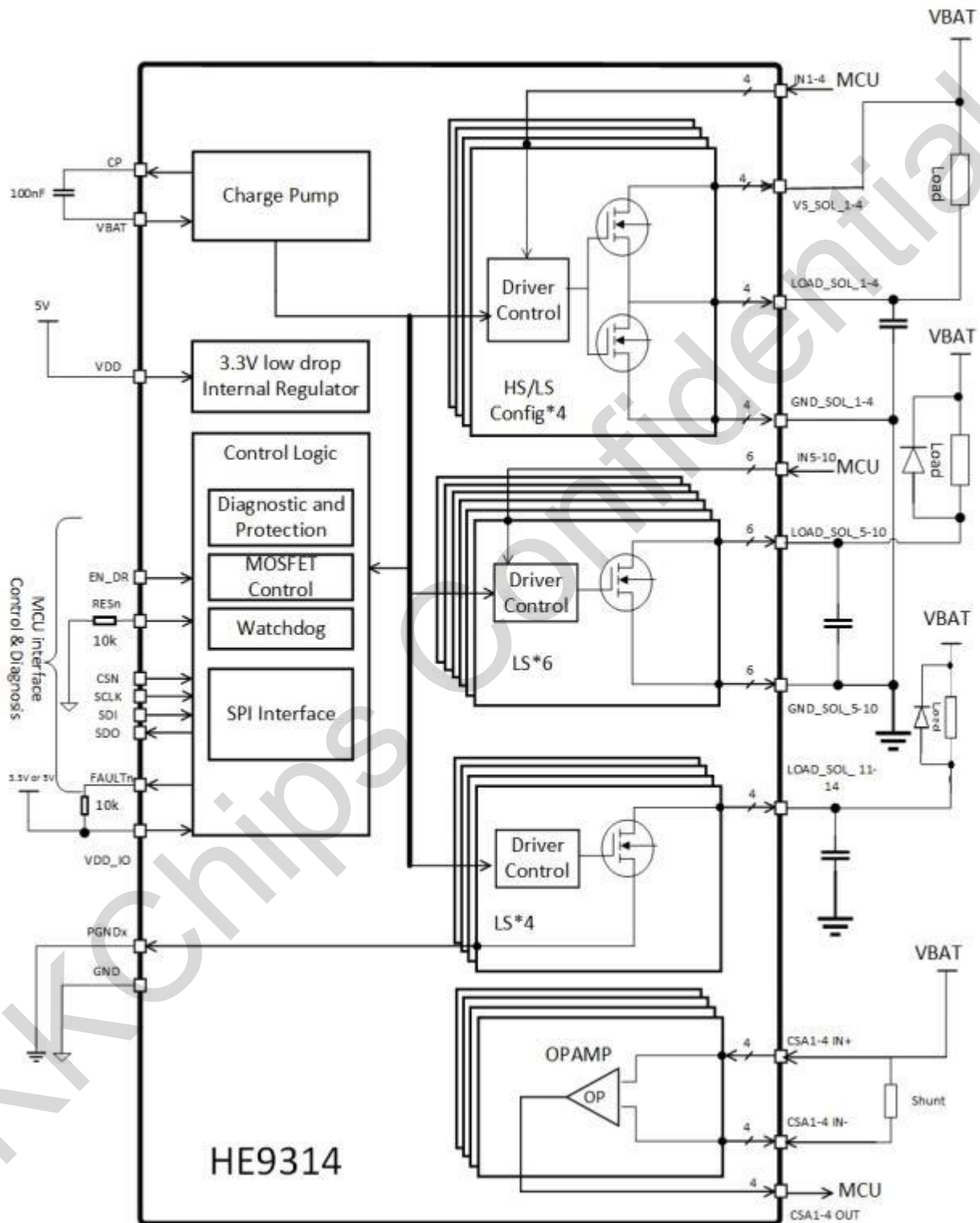


Figure 1-2 Application schematic

2 Pin configuration

2.1 Pin assignment

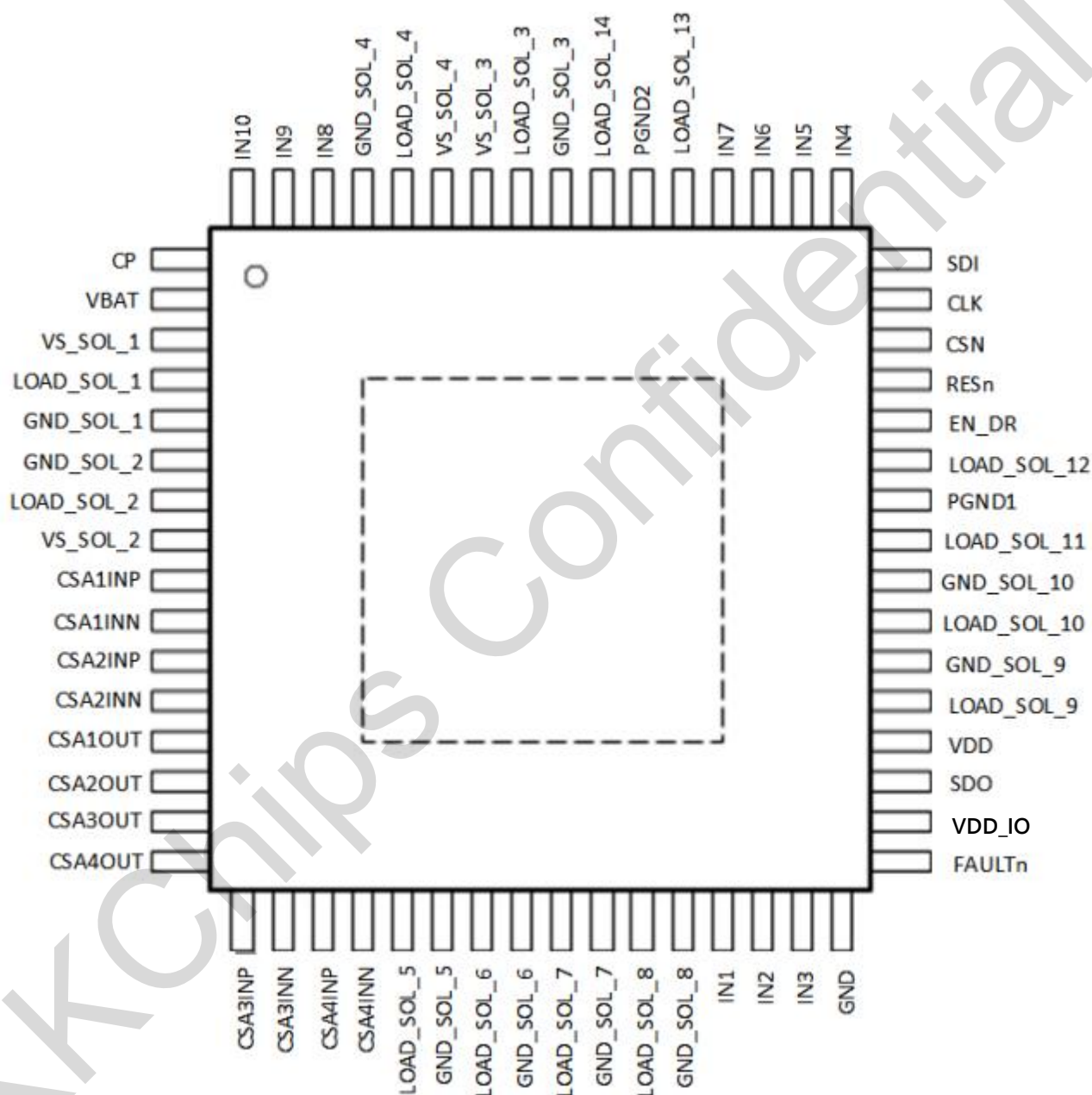


Figure 2-1 ETQFP Package

Table 2-1 Pin description

Pin#	Pin name	Function	Note
1	CP	Charge pump	
2	VBAT	Battery supply voltage	
3	VS_SOL_1	Voltage supply pin of configurable driver #1	
4	LOAD_SOL_1	Load connection pin of configurable driver #1	
5	GND_SOL_1	Ground pin of configurable driver #1	
6	GND_SOL_2	Ground pin of configurable driver #2	
7	LOAD_SOL_2	Load connection pin of configurable driver #2	
8	VS_SOL_2	Voltage supply pin of configurable driver #2	
9	CSA1INP	Current-sense amplifier positive input #1	
10	CSA1INN	Current-sense amplifier negative input #1	
11	CSA2INP	Current-sense amplifier positive input #2	
12	CSA2INN	Current-sense amplifier negative input #2	
13	CSA1OUT	Current-sense amplifier output voltage #1	
14	CSA2OUT	Current-sense amplifier output voltage #2	
15	CSA3OUT	Current-sense amplifier output voltage #3	
16	CSA4OUT	Current-sense amplifier output voltage #4	
17	CSA3INP	Current-sense amplifier positive input #3	
18	CSA3INN	Current-sense amplifier negative input #3	
19	CSA4INP	Current-sense amplifier positive input #4	
20	CSA4INN	Current-sense amplifier negative input #4	
21	LOAD_SOL_5	Load connection pin of driver #5	
22	GND_SOL_5	Ground pin of driver #5	
23	LOAD_SOL_6	Load connection pin of driver #6	
24	GND_SOL_6	Ground pin of driver #6	
25	LOAD_SOL_7	Load connection pin of driver #7	
26	GND_SOL_7	Ground pin of driver #7	
27	LOAD_SOL_8	Load connection pin of driver #8	
28	GND_SOL_8	Ground pin of driver #8	
29	IN1	Discrete input used to PWM output driver #1	
30	IN2	Discrete input used to PWM output driver #2	
31	IN3	Discrete input used to PWM output driver #3	
32	GND	Ground for analog and digital circuits	
33	FAULTn	Fault: when LOW, Fault present, when HIGH, no fault present	
34	VDD_IO	IO Supply: To be connected to 5.0V or 3.3V	
35	SDO	SPI data out	
36	VDD	Supply Voltage: Supply for circuits. Connect to 5.0V supply voltage	
37	LOAD_SOL_9	Load connection pin of driver #9	
38	GND_SOL_9	Ground pin of driver #9	

HE9314

Automotive 14 channels valve driver



Pin configuration

39	LOAD_SOL_10	Load connection pin of driver #10	
40	GND_SOL_10	Ground pin of driver #10	
41	LOAD_SOL_11	Load connection pin of driver #11	
42	PGND1	Ground pin of Power 1 (driver #11 、 #12)	
43	LOAD_SOL_12	Load connection pin of driver #12	
44	EN_DR	Enable Driver	
45	RESn	Reset input	
46	CSN	SPI chip select (active low)	
47	CLK	SPI serial clock input	
48	SDI	SPI data in	
49	IN4	Discrete input used to PWM output driver #4	
50	IN5	Discrete input used to PWM output driver #5	
51	IN6	Discrete input used to PWM output driver #6	
52	IN7	Discrete input used to PWM output driver #7	
53	LOAD_SOL_13	Load connection pin of driver #13	
54	PGND2	Ground pin of Power 2 (driver #13 、 #14)	
55	LOAD_SOL_14	Load connection pin of driver #14	
56	GND_SOL_3	Ground pin of configurable driver #3	
57	LOAD_SOL_3	Load connection pin of configurable driver #3	
58	VS_SOL_3	Voltage supply pin of configurable driver #3	
59	VS_SOL_4	Voltage supply pin of configurable driver #4	
60	LOAD_SOL_4	Load connection pin of configurable driver #4	
61	GND_SOL_4	Ground pin of configurable driver #4	
62	IN8	Discrete input used to PWM output driver #8	
63	IN9	Discrete input used to PWM output driver #9	
64	IN10	Discrete input used to PWM output driver #10	

3 Electrical specifications

3.1 Absolute maximum ratings

This part may be irreparably damaged if taken outside the values specified in the next table. Operation above the absolute maximum ratings may also cause a decrease in reliability. The operating junction temperature range is from -40°C to $+175^{\circ}\text{C}$. The maximum junction temperature must not be exceeded. All voltages are with respect to analog ground pin GND.

Table 3-1 Absolute maximum ratings

Pin name	Type	Test condition	Value			Unit	Pin type	Number
			Min	Typ	Max			
VBAT	Global	-	-0.3	-	40	V	S	P_3.1.1
CP	Local	-	VBAT-0.3	-	VB+CP_delta	V	S	P_3.1.2
VDD	Local	-	-0.3	-	20	V	S	P_3.1.3
VS_SOL_x	Global	-	-0.3	-	40 ⁽¹⁾	V	S	P_3.1.4
LOAD_SOL_x	Global	-	-0.3	-	40 ⁽¹⁾	V	O	P_3.1.5
GND_SOL_x	Global	-	-0.3	-	0.3	V	I	P_3.1.6
VDDIO	Local	-	-0.3	-	20	V	S	P_3.1.7
GND	Local	-	-0.3	-	0.3	V	S	P_3.1.8
CSN, SCLK, SDI, EN_DR, RESn, INx	Local	-	-0.3	-	20	V	I	P_3.1.9
SDO, FAULT	Local	-	-0.3	-	20	V	O	P_3.1.10
CSAxIN+, CSAxIN-	Local	-	-0.3	-	40	V	I	P_3.1.11
CSAxOUT	Local	-	-0.3	-	20	V	O	P_3.1.12

Note:

(1) 35V AMR over life-time, 40V for ISO-pulse transients (as defined in ISO7637-2 standard).

3.2 Latch up trials

Latch-up tests performed according to JEDEC 78 class 2 Level A

3.3 ESD performance

Table 3-2 ESD performance

Symbol	Parameter	Test condition	Value			Unit	Pin	Number
			Min	Typ	Max			
ESD HBM	HBM Global pins	ESD according to Human Body Model (HBM), Q100-002 for all pins; (100pF/1,5kΩ)	-4	-	4	kV	Global	
ESD HBM	HBM all pins	ESD according to Human Body Model (HBM), Q100-002 for all pins; (100pF/1,5kΩ)	-2	-	2	kV	ALL	
ESD CDM Corner	CDM corner pins	ESD according to Charged Device Model (CDM), Q100-011 Corner pins	-750	-	750	V	Corner	
ESD CDM	CDM all pins	ESD according to Charged Device Model (CDM), Q100-011 All pins	-500	-	500	V	ALL	

3.4 Temperature range

Table 3-3 Device operating temperature

Symbol	Parameter	Test condition	Value			Unit	Pin	Number
			Min	Typ	Max			
T _{amb}	Operating ambient temperature		-40	-	125	°C	ALL	
T _j	Junction temperature		-40	-	175	°C	ALL	
T _{stg}	Storage temperature	-	-50	-	175	°C	ALL	
T _{pas}	Maximum non/operating temperature during passive lifetime	-	-55	-	150 ⁽¹⁾	°C	ALL	
R _{Th j-a}	Thermal Resistance junction to ambient	Package: ETQFP-pad: 10X10X1.0mm) 2s2p (4L) board ⁽²⁾	-	20	-	°C/W	ALL	
R _{Th j-c}	Thermal Resistance junction to case	Natural convection	-	2	-			

(1) 175 °C are allowed for limited time. Eventual Mission profiles with passive lifetime temperature >150 °C have to be evaluated by KKchips to confirm that product qualification is able to cover the required conditions.

(2) JESD51-7

All parameters are guaranteed, and tested, in the temperature range from T_j -40 to 150 °C (unless otherwise specified); the device is still operative and functional at higher temperatures (up to T_j 175 °C). Device functionality at high temperature up to T_j = 175 °C is guaranteed by bench validation, electrical parameters are guaranteed by correlation with ATE tests at reduced temperature and adjusted limits (if needed).

4 Block description and electrical characteristics

4.1 Reset (RESn)

The RESn pin is the reset input for the device. If the RESn pin is low, the device is held in an internal reset state, all outputs channels are disabled, the FAULTn pin is held low and all registers are reset to their default values. An internal pull down current source will hold the RESn pin low in case the pin is open. RESn pin status is echoed in each SPI frame.

4.2 Enable (EN_DR)

The EN_DR pin is used to enable / disable the output stages. When EN_DR pin is low, all channels are disabled, and the FAULTn pin is pulled low if EN_DR is not masked. The EN_DR pin can be connected to a general purpose output pin of the microcontroller or to an alternative safety circuit.

When EN_DR pin is set high, all channels are enabled based on their configuration settings .

4.3 Channel inputs (INx)

INx pin serves as the PWM input signal to directly control the output drivers. For channels 1-4, When INx is high, the driving transistor is enabled and the active recirculation transistor is disabled. When INx is low, the driving transistor is disabled and the active recirculation transistor is enabled. The driving transistor and active recirculation transistor are dependent on the programmed output configuration, high side or low side. For channels 5-10, When INx is high, the driving transistor is enabled, When INx is low, the driving transistor is disabled.

4.4 Faultn

The device sets the FAULTn pin low upon detecting internal and external faults and while the device is in reset. The FAULTn pin is open drain and requires an external pull-up resistor connected between VDDIO and the FAULTn pin. A summary of all faults linked to the state of FAULTn are shown in [Table 4-1](#) and [Table 4-2](#). Only faults with specified masking register bits can be masked or unmasked by the user. Faults denoted as "None" are automatically linked to FAULTn output status. In addition to the items below, RESn is not masked and is linked to the FAULTn pin by default.

Table 4-1 Global and external supply fault pin error masking

FAULTn Masking function	Register
SPI Timeout Latch Fault	Servfltmask
SPI Timeout Fault	Servfltmask
VDD Under Voltage Fault	Servfltmask
VDD Over Voltage Fault	None
V3V3A Over Voltage	None
V3V3D Under Voltage	None
V3V3D Over Voltage	None
Oscillator Stuck	None
Oscillator Mismatch	None
CP Under Voltage Fault	Servfltmask
VBAT Over Voltage Fault	Servfltmask
VBAT Under Voltage Fault	Servfltmask
Core Over Temperature Fault	Servfltmask
SPI Protocol Error Fault	Servfltmask
EN_DR Echo Fault	Servfltmask
WD_reset_Fault	Servfltmask

Table 4-2 Solenoid channel error masking

FAULTn Masking function	Register & Data Bit
E_Sol_Thermal Warning	dervfltmask
PWM check fault	dervfltmask
E_Sol_Open Load Fault	dervfltmask
E_Sol_Short Fault	dervfltmask
E_LSCLAMP Fault	dervfltmask
E_Sol_HS Over Current Fault	dervfltmask
E_Sol_LS Over Current Fault	dervfltmask
E_Sol Over Temperature Fault	dervfltmask
GND_SOL Loss Fault	dervfltmask

4.5 SPI communications (CSN, SCLK, SDI & SDO)

The CSN, SCLK, SDI & SDO pins provide serial communications between the device and the microcontroller. Please see SPI for details on SPI features, device register functions and electrical characteristics.

4.6 Input / output electrical specifications

$4.75\text{ V} \leq V_{DD} \leq 5.5\text{ V}$; $5.5 \leq V_{BAT} \leq 19\text{ V}$; $-40\text{ }^{\circ}\text{C} \leq T_j \leq 175\text{ }^{\circ}\text{C}$ unless otherwise noted. All voltages refer to GND pin.

Table 4-3 Digital input/output electrical performance

Symbol	Parameter	Test condition	Min	Typ	Max	Unit	Pin	Number
EN_DR_low	Logic input low detection	–	–	–	0.75	–	EN_DR	P_4.6.1
EN_DR_high	Logic input high detection	–	1.75	–	–	V	EN_DR	P_4.6.2
EN_DR_hyst	Input hysteresis voltage	–	0.1	–	–	V	EN_DR	P_4.6.3
EN_DR_pd	Inter pull down resistance	$V_{IN} < 2.5\text{V}$	10	38	70	k Ω	EN_DR	P_4.6.4
EN_DR_deglitch	EN_DR deglitch filter	Guaranteed by scan	9	–	12	μs	EN_DR	P_4.6.5
INx_low	Logic Input low detection	–	–	–	0.75	V	INx	P_4.6.6
INx_high	Logic input high detection	–	1.75	–	–	V	INx	P_4.6.7
INx_hyst	Input hysteresis voltage	–	0.1	–	–	V	INx	P_4.6.8
INx_pd	Inter pull down resistance	$V_{IN} < 2.5\text{V}$	10	38	70	k Ω	INx	P_4.6.9
RESn_low	Logic input low detection	–	–	–	0.75	–	RES	P_4.6.10
RESn_high	Logic input high detection	–	1.75	–	–	V	RES	P_4.6.11
RESn_hyst	Input hysteresis voltage	–	0.1	–	–	V	RES	P_4.6.12
RESn_pd	Inter pull down resistance	$V_{IN} < 2.5\text{V}$	10	38	70	k Ω	RES	P_4.6.13
RESn_deglitch	RESn deglitch filter	Guaranteed by scan	4.5	5	7	μs	RES	P_4.6.14
FAULTn_ileak	Leakage current in tri-state	–	–	–	10	–	FAULT	P_4.6.15

Block description and electrical characteristics

	condition							
FAULTn_low	Logic low level	Ext pull-up 4.7k to VDDIO	–	–	0.4	–	FAULT	P_4.6.16
FAULTn_high	Logic high level	Ext pull-up 4.7k to VDDIO	VDDIO-0.4	–	–	V	FAULT	P_4.6.17
FAULTn_trise	FAULTn rise time	Ext pull-up 4.7k to VDDIO, Cload=20pF, tested by correlation	–	–	–	ns	FAULT	P_4.6.18
FAULTn_tfall	FAULTn fall time	Ext pull-up 4.7k to VDDIO, Cload=20pF, tested by correlation	–	–	–	ns	FAULT	P_4.6.19

5 Power supply

5.1 Overview

The HE9314 has multiple supply pins. Internal circuits are powered from VDD (+5 V) and VBAT (reverse battery protected) while digital I/O pins are powered from VDDIO (3.3 V or 5 V) for consistency with the main microcontroller I/O operation. The charge pump regulator is sourced from VBAT and supplies the necessary voltage source for the high side integrated valve drivers.

5.2 Battery supply (VBAT)

This pin is the supply for the charge pump and must be connected to the reverse polarity protected battery voltage supply. VBAT input voltage is diagnosed for over voltage conditions. The status of this diagnostic is readable by SPI. Refer to [Section 7.1.2](#) VBAT supply over voltage for additional information. The following table summarizes the functional ranges dependent on battery supply voltage.

Table 5-1 VBAT electrical performance

Symbol	Parameter	Test condition	Min	Typ	Max	Unit	Number
VBAT operational	Normal operating voltage range ⁽¹⁾	Tamb = -40 °C , 4.75 V ≤ VDD ≤ 5.5 V	5.5	–	19	V	
		Tamb = 27 °C , 4.75 V ≤ VDD ≤ 5.5 V	5.55	–	19	V	
		Tamb = 125 °C , 4.75 V ≤ VDD ≤ 5.5 V	5.6	–	19	V	
VBAT double/load dump	High voltage range ⁽²⁾	Tamb ≤ 50 °C , 4.75 V ≤ VDD ≤ 5.5 V	19	–	36	V	
VBAT over voltage	Over voltage range ⁽³⁾	Tamb ≤ 50 °C , 4.75 V ≤ VDD ≤ 5.5 V	36	–	40	V	
VBAT low voltage	Low Voltage Range Cranking ⁽⁴⁾	Tamb = -40 °C , 4.75 V ≤ VDD ≤ 5.5 V	–	–	5.5	V	
		Tamb = 27 °C , 4.75 V ≤ VDD ≤ 5.5 V	–	–	5.55	V	
		Tamb = 125 °C , 4.75 V ≤ VDD ≤ 5.5 V	–	–	5.6	V	

1. Device is capable of full functional operation; VBAT must be reversed battery protected
2. Device is capable of full functional operation at Ta ≤ 50 °C with possible degradation of electrical parameters linked to battery line (ex. solenoid slew rate control); There will be no

damage to the device, no false operation and input pins will withstand voltage and current defined in Absolute ratings regardless of battery voltage range. Full functional operation will resume without operator intervention when battery voltage returns to Normal Operating Voltage Range at $T_a \leq 50^\circ\text{C}$.

3. $T_a \leq 50^\circ\text{C}$, duration less than 500ms. There will be no damage to the device, Solenoid channels will be disabled and input pins will withstand voltages and currents defined in Absolute ratings regardless of battery voltage range. Full functional operation will resume with operator intervention, as described in the specification, when battery voltage returns to normal operating voltage range.
4. Device is capable of full functional operation until charge pump is out of under voltage condition with possible degradation of electrical parameters linked to battery line (ex. solenoid slew rate control). Full functional operation will resume without operator intervention when battery voltage returns to normal operating voltage range.

5.3 Load supplies (VSOL1, VSOL2, VSOL3 & VSOL4)

The VSOLx pins are connected to the high side of the solenoid load when the output drivers are configured for low side operation. In low side operation, VSOLx provides power to the load and recirculation path through the high side driver. When the output drivers are configured for high side operation, the VSOLx pin becomes the power source for driving the load thru the high side solenoid driver transistor. In this case, load current is recirculated through the corresponding low side driver. VSOLx pins are independent for each solenoid driver channel.

5.4 Analog and digital supply (VDD)

The VDD pin provides the source voltage for all internal analog (V3V3A) and digital (V3V3D) circuits. An internal start-up circuit connected to VDD initializes the internal regulators (V3V3A & V3V3D) once VDD reaches VDD_UV threshold. V3V3A & V3V3D regulators are monitored for internal power on reset control (POR). VDD input voltage is diagnosed for over and under voltage conditions. The status of this diagnostic is readable by SPI. Please see [Section 7.1.1](#) VDD over and under voltage diagnostics.

5.5 I/O supply (VDDIO)

VDDIO is the supply for all pins that interface with the external microcontroller. This pin must be connected to a supply with the same voltage used by the microcontroller I/O, 3.3 V or 5.0 V. There is no monitoring function associated with VDDIO.

5.6 Power-On reset

An internal power on reset circuit holds the device in a reset state if VDD goes below the under voltage threshold. The power on reset is released once VDD is within normal operating range for the specified reset filter time TPOR. The SPI interface can be accessed after the power on reset time.

The fault bit "POR_flag" is set, when the device exits the reset state. This bit is cleared automatically, whenever the register is accessed. Also res_echo in PINSTATUS register is set, when

the device exits the reset state in case of Power-On reset event, despite the state of RESn pin. The microcontroller can use these bits to determine if an internal or external reset has occurred.

A power on reset (POR) can also occur if any of the internal bandgap regulators or internal analog & digital regulators are diagnosed as over or under voltage. Additional details on these and VDD monitoring can be found in sections [Section 7.1.1](#) and [Section 7.1.4](#).

5.7 GND (GND, PGND1/2, GND_SOLx)

GND is used to connect internal analog and digital control circuits. Exposed pad is connected to substrate and to GND. The device has 12 power ground connections, GND_SOL0-10, PGND1/2. All ground pins have open circuit detection. Additional details on ground loss detection are shown in [Section 7.1.5](#) GND loss detection.

5.8 Charge pump

On the CP pin it's required to connect a 100nF capacitor toward the VB line. The charge pump is ON if $VDD > UV$ threshold

If $VDD < V3V3$, the voltage of the charge pump begins to decrease. The final CP voltage is equal to $VB - V_{be}$.

When the CP is ON, the low CP diagnosis is enabled. When any HS is switched on but CP voltage is not high enough to switch on HS, low CP fault is detected and stored in cp SPI flag.

$5.5\text{ V} \leq VB < 19\text{ V}$; $-40\text{ }^{\circ}\text{C} \leq T_j \leq 175\text{ }^{\circ}\text{C}$ unless otherwise specified.

Table 5-2 charge pump electrical performance

Pin	Symbol	Parameter	Test condition	Min.	Typ.	Max.	Unit	Number
CP	CP_DELTA	Delta voltage CP-VB	$VB > 5V$	3	-	6	V	P_5.8.1

5.9 Startup & power down sequence

HE9314 begins initialization and execution of all internal diagnostic and self-test functions upon VDD reaching the VDD_UV threshold. A detailed list of these monitoring functions is summarized in [Section 7](#) Safety features.

While the device is initializing, the FAULn pin remains low. Upon completing initialization, the SPI POR flag is set high (clear on read) to indicate a POR event has occurred and the FAULn pin remains low until the RES input is set high. The RES input can be set high prior to the device completing initialization but the FAULn pin will not be released until the POR event is complete. This sequence is shown in [Figure 5-1](#). Also RES_echo in PINSTATUS register is set, when the device exits the reset state in case of Power-On reset event, despite the state of RESn pin.

Once the POR event is complete and the RESn input is set high, the device can be configured by the

user. All registers maintain RESET default states until both POR is set high and RES is set by the user.

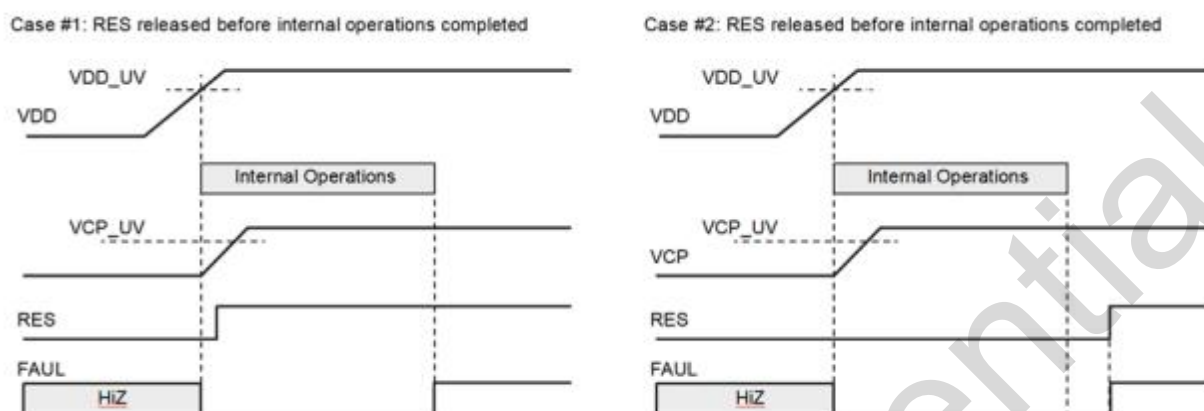


Figure 5-1 Startup procedure example

HE9314 power down sequence requires only the VDD supply be disabled. Once the VDD_UV threshold is reached, the solenoid are disabled. There are no timing dependencies between VDD and VBAT voltage levels to perform the power down sequence.

5.10 Power supply electrical specifications

$4.75V \leq VDD \leq 5.5V$; $5.5 \leq VBAT \leq 19V$; $-40^{\circ}C \leq T_j \leq 175^{\circ}C$ unless otherwise noted. All voltages refer to GND pin.

Table 5-3 Power supply electrical performance

Symbol	Parameter	Test condition	Min	Typ	Max	Unit	Pin	Number
POR_deglitch	POR analog deglitch		5	18	35	μs	POR	P_5.10.1
VBAT_stby_cur	VBAT stby current consumption	VDD = 0 V, VBAT = 13 V	–	–	45	μA	VBAT	P_5.10.2
VDD_stby_cur	VDD stby current consumption	VDD < 0.2 V	–	–	2	μA	VDD	P_5.10.3
VBAT_on_state_cur	VBAT on state current consumption	VDD = 5 V, VBAT = 13 V, All channels tristate	5	10	20	mA	VBAT	P_5.10.4
VDD_on_state_cur	VDD on state current consumption	VDD = 5 V	20	30	40	mA	VDD	P_5.10.5
VDDIO_stby_curr	VDDIO stby current consumption	VDD = 0 V, VDDIO = 5 V	-1	–	10	μA	VDDIO	P_5.10.6
VDDIO_on_state_curr	VDDIO on state current consumption	VDD = 5 V, VDDIO = 5 V	–	–	5	mA	VDDIO	P_5.10.7

Power supply

V3V3A	Internal analog supply line	Design info, not tested	–	3.3	–	V	V3V3A	P_5.10.8
V3V3D	Internal digital supply line	Design info, not tested	–	3.3	–	V	V3V3D	P_5.10.9
fosc_main	Main oscillator frequency	–	-5%	16	5%	MHz	fosc	P_5.10.10
fosc_aux	Auxiliary oscillator frequency	–	-5%	16	5%	MHz	fosc_aux	P_5.10.11
fosc_mod_freq	Spread spectrum modulation frequency	Design info, not tested	–	125	–	kHz	fosc	P_5.10.12
fosc_mod_index_min	Spread spectrum minimum modulation index		-5	–	-2	%	fosc	P_5.10.13
fosc_mod_index_max	Spread spectrum maximum modulation index		2	–	5	%	fosc	P_5.10.14
tasic_init	Initialization time of ASIC	Time from VDD rising to FAUL release, RES HIGH	–	5	10	ms	VDD, FAUL	P_5.10.15

6 SOL_driver

6.1 SOL_driver_1-4

The HE9314 provides four independent current control channels. Each channel includes the Driver Control Block and Output Stage. The Output Stage consists of a driving transistor and the recirculation transistor and is user configurable for high side or low side control. When configured as a low side driver, the high side driver recirculates load current. When configured as a high side driver, the low side driver recirculates load current. Configuration examples are shown in [Figure 6-1](#).

The high or low side configuration is selected in register, solenoid driver configuration, The device defaults to low side configuration.

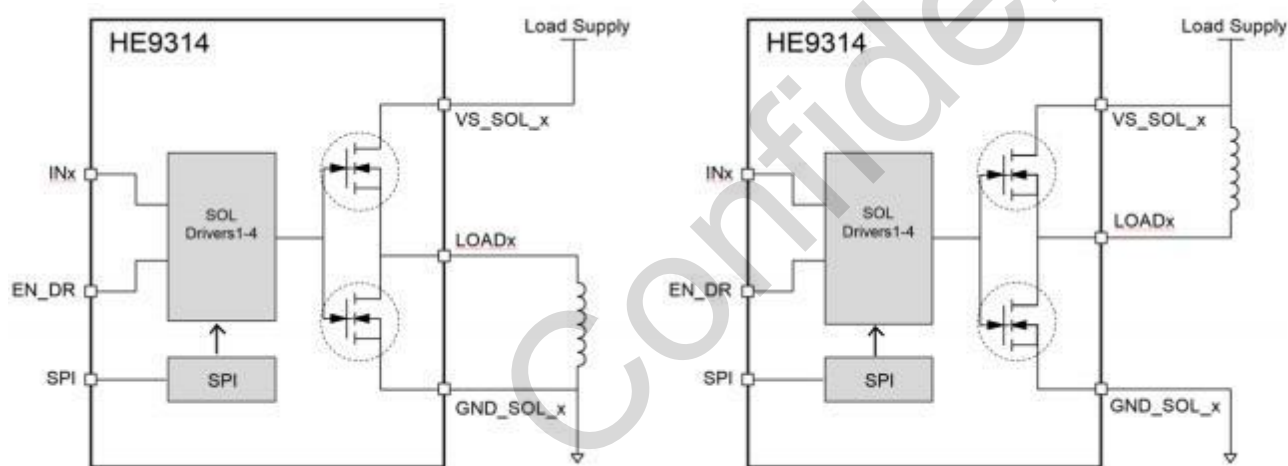


Figure 6-1 HS driver (left) and LS driver (right) configuration

6.2 SOL_driver_5-14

SOL_Drive_5-10 is controlled by INx and SPI, SOL_drive_11-14 is controlled by spi.

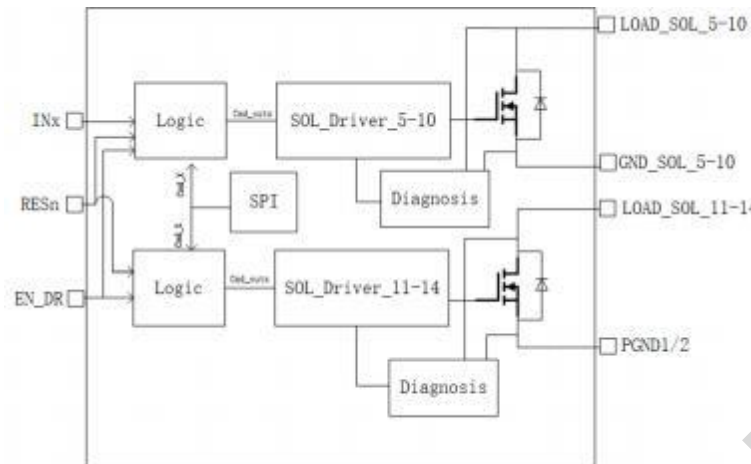


Figure 6-2 SOL_driver_5-14 diagram

6.3 Parallel mode

There are five couples of drivers in parallel to get five channels with lower RdsON: LOAD_SOL_1 with LOAD_SOL_2, LOAD_SOL_3 with LOAD_SOL_4, LOAD_SOL_5 with LOAD_SOL_6, LOAD_SOL_7 with LOAD_SOL_8, LOAD_SOL_9 with LOAD_SOL_10, LOAD_SOL_11 with LOAD_SOL_12, LOAD_SOL_13 with LOAD_SOL_14.

To enable parallel mode the following conditions must be verified, otherwise the channels will be kept independent

depending on their own settings:

Bit [15:9] in solenoid channel parallel mode configuration register must be set to 1 :

Bit [15] enables parallel mode for LOAD_SOL_13 and LOAD_SOL_14;

Bit [14] enables parallel mode for LOAD_SOL_11 and LOAD_SOL_12;

Bit [13] enables parallel mode for LOAD_SOL_9 and LOAD_SOL_10;

Bit [12] enables parallel mode for LOAD_SOL_7 and LOAD_SOL_8;

Bit [11] enables parallel mode for LOAD_SOL_5 and LOAD_SOL_6;

Bit [10] enables parallel mode for LOAD_SOL_3 and LOAD_SOL_4;

Bit [9] enables parallel mode for LOAD_SOL_1 and LOAD_SOL_2;

6.4 PWM control

To avoid spurious switching of the external loads due to noise spikes on INx pins, a digital filter, LOADx_ton_res and LOADx_toff_res (where μs is related to selected channel), has been added to these pins.

Since the filter time is applied on each transition of the INx pin signal, OFF to ON and ON to OFF, the effect limits the minimum period or maximum frequency of the signal delivered to the LOADx pin.

Channel state depends on the EN_DR input pin and the value of the Driver Configuration/Status SPI register .

Table 6-1 INx polarity

Driver Configuration	Driver Status	INx Pin Status	
		HIGH	LOW
LOW Side (channel 1-4) (load connected between VS_SOLx pin and LOAD_SOLx pin)	Low side FET	ON	OFF
	High side FET	OFF	ON
HIGH Side (channel 1-4) (load connected between LOAD_SOLx pin and GND_SOLx pin)	Low side FET	OFF	ON
	High side FET	ON	OFF
LOW Side (channel 5-10)	Low side FET	ON	OFF

FULL_ON Mode: Driver is supplied and in FULL_ON mode. In this condition, the driver is kept on continuously, until the FULL_ON mode is removed or a fault occurs.

6.4.1 Frequency Modulation

PWM mode and frequency modulation is activated with the PWM_EN bit. The band and speed is defined by four SPI bits.

Four bits for the frequency modulation band (FM_LSD_MB[3:0]): the number of frequency steps

Two bits for the frequency modulation period (FM_LSD_MP[1:0]): the number of PWM periods between frequency steps

Table 6-2 PWM Control Electrical Characteristics (SOL_Driver 1-14)

DUTY_VLV	Duty cycle Programming (8bits) x_DC[7:0] bits	—		—	
	0000 0000	—	0	—	
	...	—	—	—	
	0110 0100	—	50	—	%
	...	—	—	—	
	1100 1000	—	100	—	
	1111 1111	—	100	—	
fLF_PWM	Output PWM Frequency for LSDx (4-bits) LF_PWM[3:0] bits				
	0000		1.0		
	0001		1.5		
	0010		2.0		
	0011		2.5		
	0100		3.5		
	0101		4.0		
	0110		4.5		
	0111	-12%	8.0	+12%	kHz
	1000		9.5		
	1001		10.0		
	1010		10.5		
	1011		15.5		
	1100		16.0		
	1101		16.5		
	1110		18		
	1111		20		
FM_LSD_MB[3:0] ₍₁₎	Four SPI bits for Frequency Modulation Band	0000 = no modulation (default) 0001 = 0.5%/1 step 0010 = 1%/2 step ... 1111 = 7.5%/15 step			
FM_LSD_MP[1:0] ₍₁₎	Two SPI bits for Frequency Modulation Period	00 = 8 cycle 01 = 16 cycle 10 = 32 cycle (default) 11 = 64 cycle			

Note: 1: PWM mode active and Frequency modulation active if pwm_enable bit = 1

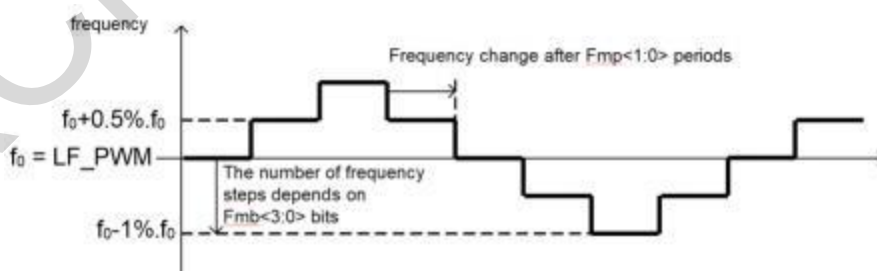


Figure 6-3 Frequency Modulation Description (2 Steps)

6.5 Solenoid driver and load diagnostic

6.5.1 Thermal protection

Each solenoid channel has a dedicated temperature sensor that continuously monitors the temperature of the Low Side and of the High Side power transistor. Temperature Diagnostic information is available in the dedicated SPI register Temperature Diagnostics.

Should channel temperature reach the thermal warning threshold T_WARN , the thermal warning flag is set in the Temperature register. The warning flag is latched and is cleared upon SPI read if the thermal warning condition is no longer present.

Should channel temperature continue to rise up to the thermal shutdown threshold T_SD , the related channel is disable. This event has a dedicated fault flag in the Temperature register,. Channel activation is prevented if thermal shut down condition is still present on SPI read. This fault is latched and is cleared upon.

6.5.2 Overcurrent protection

Overcurrent protection is present for each HS and LS solenoid driver. The overcurrent threshold is selectable through SPI register solenoid driver configuration

Should an overcurrent fault be detected (on HS or on LS), the related channel is disable and a dedicated diagnostic bit is set in register exception .To restart the channel, the MCU must clear the fault bit by reading the fault register and re-enable the channel.

6.5.3 Supply disconnection

When a solenoid channel is configured for low side operation, current recirculation occurs through the internal high side transistor. If the supply Vs_solx of the high side driver is lost, the voltage at $LOAD_SOLx$ pin can become unpredictable and possibly exceed the absolute rating of the device. To prevent this condition from damaging the device, the low side driver has an integrated clamping circuit to limit the peak voltage.

Should the low side clamping structure be activated during normal operation (i.e. solenoid supply line within normal operating range), the channel is disable and the Low Side Clamp Active fault flag is set in the exceptions register. To restart the channel, the MCU must clear the fault by reading the exceptions register and re-enable the channel. The clamp flag should be activated after enable the off diagnostic , but this will not affect the hardware function of the clamp.

Activation of LS clamp during a load dump event could damage the device due to the high energy of load dump event. To avoid this condition, the LS clamp circuitry is automatically disabled when VBAT pin exceeds deep over-voltage threshold.

6.5.4 Off state solenoid diagnostics [sol_driver_1-4]

The device provides independent off-state diagnostics for each channel. Simplified diagrams are shown in [Figure 6-4](#) and [Figure 6-5](#) below, depending on the chosen configuration.

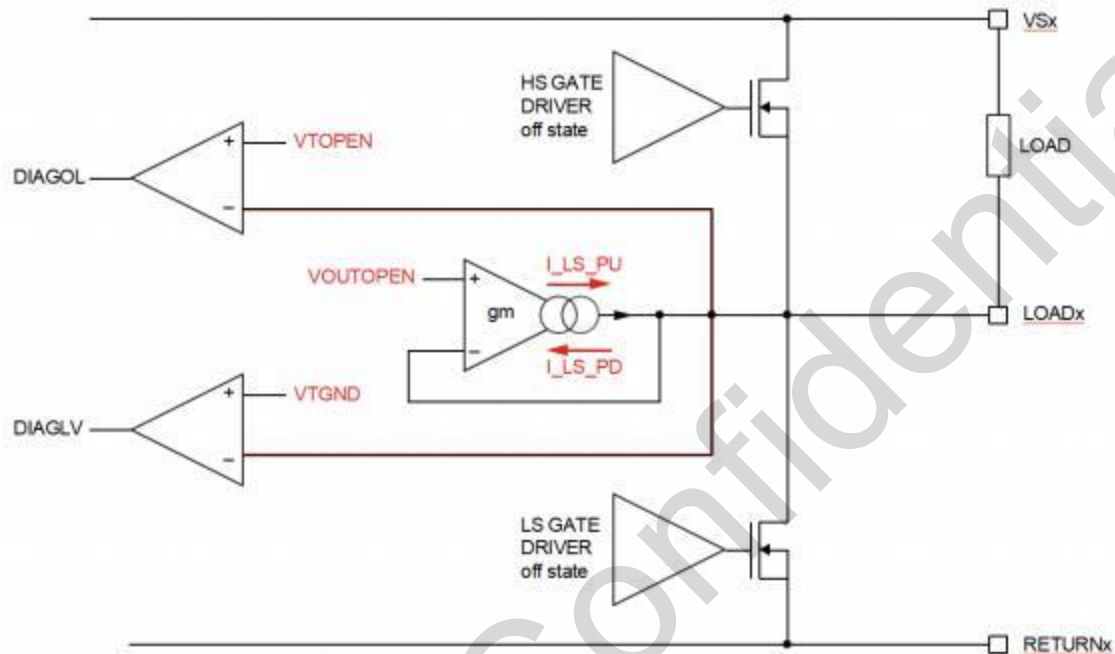


Figure 6-4 Low side configuration diagnostic

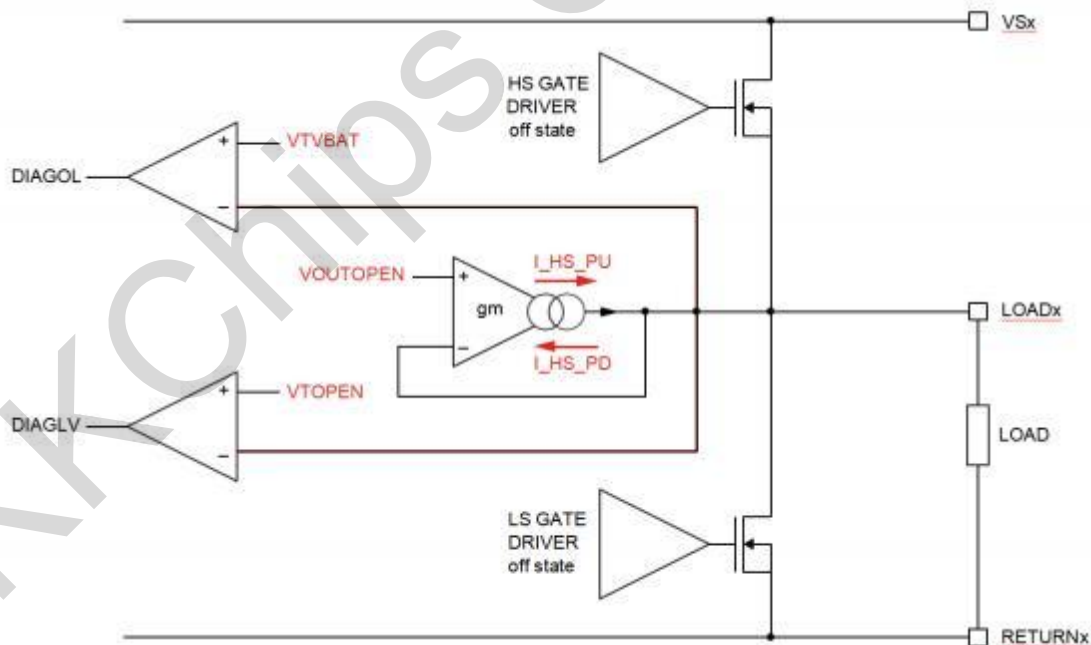


Figure 6-5 High side configuration diagnostic

By default, diagnostic pull up/down current sources are disabled and comparator outputs are

masked by internal logic. To enable OFF state diagnostic, the channel must be set the OFF diagnostic. In case the desired channel is put in tristate after operation a false diagnostic can be sensed due to the time needed to discharge output voltage through the load. To avoid incorrect fault detection, the fault comparators are masked for t_{d_blank} filter time. The diagnostic blanking time is programmable through SPI register solenoid driver configuration. Blanking time is programmable to best align with external load values and ESD capacitor connected to the output pin. Fault conditions are latched and cleared on SPI read, however they do not prevent from enabling the solenoid channels. OFF state diagnostic comparators are the same for both HS and LS configuration and their output is decoded from internal logic to set the SPI fault bits exceptions according to the following tables

NOTE: To ensure the accuracy of off diagnosis, Driver_en should be set to 0 and off diagnostic set to 1 (channel 1-4).
Driver_en should be set to 1 and off diagnostic set to 1 (channel 5-14).

Table 6-3 LS diagnostic truth table

Comparator output		Fault detection	Exception registers	
DIAGOL	DIAGLV		OPEN LOAD	SHORT
0	0	Normal operation – no fault-	0	0
0	1	Not possible	0	0
1	0	Open load	1	0
1	1	Short to ground	0	1

Table 6-4 HS diagnostic truth table

Comparator output		Fault detection	Exception registers	
DIAGOL	DIAGLV		OPEN LOAD	SHORT
0	0	Short to Battery	0	1
0	1	Not possible	0	0
1	0	Open load	1	0
1	1	Normal operation – no fault	0	0

6.5.5 Off state solenoid diagnostics [sol_driver_5-14]

The device provides independent off-state diagnostics for each channel. Simplified diagrams are shown in [Figure 6-6](#) below, depending on the chosen configuration.

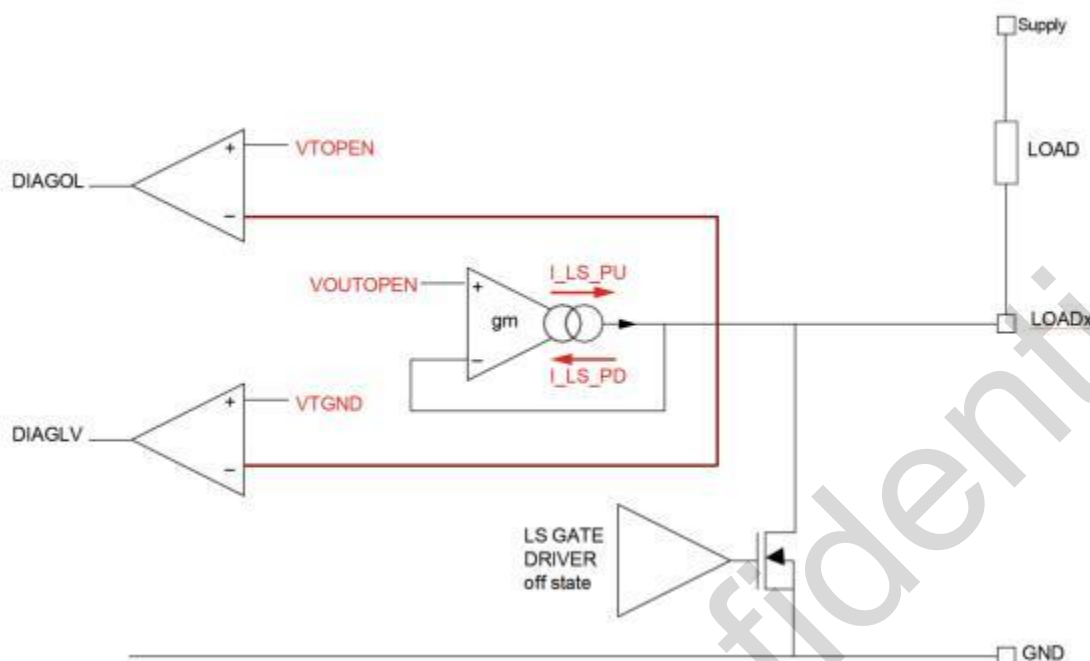


Figure 6-6 Low side diagnostic

Table 6-5 LS diagnostic truth table

Comparator output		Fault detection	Exception registers	
DIAGOL	DIAGLV		OPEN LOAD	SHORT
0	0	Normal operation – no fault-	0	0
0	1	Not possible	0	0
1	0	Open load	1	0
1	1	Short to ground	0	1

6.6 Valve driver electrical specifications (LOAD_SOL_1-4)

$4.75\text{ V} \leq V_{DD} \leq 5.5\text{ V}$; $5.5\text{ V} \leq V_{BAT} \leq 19\text{ V}$; $V_{S_SOLx} \leq 19\text{ V}$; $V_{BAT}-1\text{ V} \leq V_{S_SOLx} \leq V_{BAT}+1\text{ V}$;
 $-40\text{ }^{\circ}\text{C} \leq T_j \leq 175\text{ }^{\circ}\text{C}$ unless otherwise noted. All voltages refer to GND pin.

Table 6-6 Valve driver power stage electrical parameters

Symbol	Parameter	Test condition	Min	Typ	Max	Unit	Pin	Number
Solenoid driver power stage								
Irev_VSx	VSx reverse current	Application note: in the application	-3	–	–	A	VS_SOLx	P_6.6.1

		condition, no impact on other functions, no reset generation						
Irev_LOADx	LOADx reverse current	Application note: in the application condition, no impact on other functions, no leakage from nearby pins, no reset generation	-3	–	–	A	LOAD_SOLx	P_6.6.2
VSx_lkg	VSx current leakage stby	Output disabled, VDD = 0 V, VS_SOLx = 13 V	-1	–	1	μA	VS_SOLx	P_6.6.3
LOADx_lkg	LOADx current leakage tristate	Output disabled, VDD = 5 V, VS_SOLx = 13 V, LOAD_SOLx = VS_SOLx or GND	-140	–	100	μA	LOAD_SOLx	P_6.6.4
HS_RdsON	High side RdsON	Tj = 150 °C , ILOAD = 1 A	–	–	0.3	Ω	LOAD_SOLx	P_6.6.5
LS_RdsON	Low side RdsON	Tj = 150 °C , ILOAD = 1 A	–	–	0.3	Ω	LOAD_SOLx	P_6.6.6
LOAD_SOLx_ton_res	ton filter time	Guaranteed by scan	3	3.5	3.9	μs	LOAD_SOLx	P_6.6.7
LOAD_SOLx_toff_res	toff filter time	Guaranteed by scan	3	3.5	3.9	μs	LOAD_SOLx	P_6.6.8
LOAD_SOLx_SR00	Voltage slew rate SR00	VB = 13.5 V Load: 8 Ω, 10 nF – From 80% to 20% & 20% to 80% of OUTx	0.6	1	1.8	V/μs	LOAD_SOLx	P_6.6.9
LOAD_SOLx_SR01	Voltage slew rate SR01	VB = 13.5 V Load: 8 Ω, 10 nF – From 80% to 20% & 20% to 80% of OUTx	2	4	6	V/μs	LOAD_SOLx	P_6.6.10
LOAD_SOLx_SR10	Voltage slew rate SR10	VB = 13.5 V Load: 8 Ω, 10 nF – From 80% to 20% & 20% to 80% of OUTx	5	10	20	V/μs	LOAD_SOLx	P_6.6.11
LOAD_SOLx_SR11	Voltage slew rate SR11	VB = 13.5 V Load: 8 Ω, 10 nF – From 80% to 20% & 20% to 80% of OUTx	14	20	32	V/μs	LOAD_SOLx	P_6.6.12
LOAD_SOLx_ton00_HS	ton00_HS_mode	VB=VS_SOL=13.5 V Load: 8 Ω, 10 nF – – from INx or CS to 20% output	6	12	22	μs	LOAD_SOLx	P_6.6.13
LOAD_SOLx_ton00_LS	ton00_LS_mode	VB=VS_SOL=13.5 V Load: 8 Ω, 10 nF – – from INx or CS to 80% output	2	5	10	μs	LOAD_SOLx	P_6.6.14
LOAD_SOLx_ton01_HS	ton01_HS_mode	VB=VS_SOL=13.5 V Load: 8 Ω, 10 nF – – from INx or CS	1.4	2.8	6.5	μs	LOAD_SOLx	P_6.6.15

		to 20% output						
LOAD_SOLx_ton01_LS	ton 01_LS_mode	VB=VS_SOL=13.5 V Load: 8 Ω , 10 nF – from INx or CS to 80% output	0.8	1.8	4	μ s	LOAD_SOLx	P_6.6.16
LOAD_SOLx_ton10_HS	ton 10_HS_mode	VB=VS_SOL=13.5 V Load: 8 Ω , 10 nF – from INx or CS to 20% output	0.5	1	2.5	μ s	LOAD_SOLx	P_6.6.17
LOAD_SOLx_ton10_LS	ton 10_LS_mode	VB=VS_SOL=13.5 V Load: 8 Ω , 10 nF – from INx or CS to 80% output	0.6	1.3	2.5	μ s	LOAD_SOLx	P_6.6.18
LOAD_SOLx_ton11_HS	ton 11_HS_mode	VB=VS_SOL=13.5 V Load: 8 Ω , 10 nF – from INx or CS to 20% output	0.2	0.55	2	μ s	LOAD_SOLx	P_6.6.19
LOAD_SOLx_ton11_LS	ton 11_LS_mode	VB=VS_SOL=13.5 V Load: 8 Ω , 10 nF – from INx or CS to 80% output	0.4	1	2	μ s	LOAD_SOLx	P_6.6.20
LOAD_SOLx_toff00_HS	toff 00_HS_mode	VB=VS_SOL=13.5 V Load: 8 Ω , 10 nF – from INx or CS to 80% outout	6	12	22	μ s	LOAD_SOLx	P_6.6.21
LOAD_SOLx_toff00_LS	toff 00_LS_mode	VB=VS_SOL=13.5 V Load: 8 Ω , 10 nF – from INx or CS to 20% outout	2	5	10	μ s	LOAD_SOLx	P_6.6.22
LOAD_SOLx_toff01_HS	toff 01_HS_mode	VB=VS_SOL=13.5 V Load: 8 Ω , 10 nF – from INx or CS to 80% outout	1.4	4	6.5	μ s	LOAD_SOLx	P_6.6.23
LOAD_SOLx_toff01_LS	toff 01_LS_mode	VB=VS_SOL=13.5 V Load: 8 Ω , 10 nF – from INx or CS to 20% outout	0.8	1.8	4	μ s	LOAD_SOLx	P_6.6.24
LOAD_SOLx_toff10_HS	toff 10_HS_mode	VB=VS_SOL=13.5 V Load: 8 Ω , 10 nF – from INx or CS to 80% outout	0.5	1	2.5	μ s	LOAD_SOLx	P_6.6.25
LOAD_SOLx_toff10_LS	toff 10_LS_mode	VB=VS_SOL=13.5 V Load: 8 Ω , 10 nF – from INx or CS to 20% outout	0.6	1.3	2.5	μ s	LOAD_SOLx	P_6.6.26
LOAD_SOLx_toff11_HS	toff 11_HS_mode	VB=VS_SOL=13.5 V Load: 8 Ω , 10 nF – from INx or CS to 80% outout	0.2	0.55	1	μ s	LOAD_SOLx	P_6.6.27
LOAD_SOLx_toff11_LS	toff 11_LS_mode	VB=VS_SOL=13.5 V Load: 8 Ω , 10 nF – from INx or CS to 20% outout	0.4	1	2	μ s	LOAD_SOLx	P_6.6.28

Table 6-7 Valve driver on state diagnostics parameters

Symbol	Parameter	Test condition	Min	Typ	Max	Unit	Pin	Number
On state diagnostic								
I_HS_ovc0	HS overcurrent	OVC_thres_x=0	–	–	-3	A	LOAD_SOLx	P_6.6.29
I_HS_ovc1	HS overcurrent	OVC_thres_x=1	–	–	-4.5	A	LOAD_SOLx	P_6.6.30
I_LS_ovc0	LS overcurrent	OVC_thres_x=0	3	–	–	A	LOAD_SOLx	P_6.6.31
I_LS_ovc1	LS overcurrent	OVC_thres_x=1	4.3	–	–	A	LOAD_SOLx	P_6.6.32
Inx_fit_00	Inx time_00 filter	Guaranteed by scan		0		μs	Inx	P_6.6.33
Inx_fit_01	Inx time_01 filter	Guaranteed by scan		0.5		μs	Inx	P_6.6.34
Inx_fit_10	Inx time_10 filter	Guaranteed by scan		1		μs	Inx	P_6.6.35
Inx_fit_11	Inx time_11 filter	Guaranteed by scan		2		μs	Inx	P_6.6.36
I_HS_ovc_fit	HS overcurrent filter time	Guaranteed by scan	3	–	5	μs	LOAD_SOLx	P_6.6.37
I_LS_ovc_fit	LS overcurrent filter time	Guaranteed by scan	3	–	5	μs	LOAD_SOLx	P_6.6.38
T_WARN	Temperature warning	–	175	–	185	°C	Tj_SOLx	P_6.6.39
T_SD	Temperature shut down	–	185	–	200	°C	Tj_SOLx	P_6.6.40
T_SD_deglitch	Digital deglitch filter time	Guaranteed by scan	7.5	10	12.5	μs	Tj_SOLx	P_6.6.41
T_WARN_deglitch	Digital deglitch filter time	Guaranteed by scan	7.5	10	12.5	μs	Tj_SOLx	P_6.6.42
VBAT_ov_th1	VBAT over voltage first threshold	–	27	27.8	29	V	VBAT	P_6.6.43
VBAT_ov_th1_deglitch	VBAT over voltage digital deglitch	Guaranteed by scan	90	–	120	μs	VBAT	P_6.6.44
Vclamp	LOADx clamp voltage	VBAT < 27V	34	37	40	V	LOAD_SOLx	P_6.6.45
Vclamp_dly	LOADx disable delay by clamp activation detection	Design info, not tested	–	–	2	μs	LOAD_SOLx	

Vclamp_fit	LOADx clamp detection filter time	Guaranteed by scan	3.5	5	7	μs	LOAD_SOLx	P_6.6.47
Eclamp	Energy clamp single pulse	Iload = 1 A, Tj = 100 °C	–	–	25	mJ	LOAD_SOLx	P_6.6.48
Eclamp	Energy clamp single pulse	Iload = 1 A, Tj = 175 °C	–	–	18	mJ	LOAD_SOLx	P_6.6.49

Table 6-8 Valve driver LS off state diagnostics parameters

Symbol	Parameter	Test condition	Min	Typ	Max	Unit	Pin	Number
Off state diagnostic – LS configuration								
VTGND	Short to GND threshold voltage	LS mode, driver tristate, diagnostic on	1.9	2.1	2.3	V	LOAD_SOLx	P_6.6.50
VTOPEN	Open load threshold voltage	LS mode, driver tristate, diagnostic on	2.7	2.9	3.1	V	LOAD_SOLx	P_6.6.51
VOUOPEN	Open load voltage	LS mode, driver tristate, diagnostic on	2.35	2.5	2.65	V	LOAD_SOLx	P_6.6.52
I_LS_THOPEN	Open load threshold current	–	40	60	90	μA	LOAD_SOLx	P_6.6.53
I_LS_PD	Diagnostic pull down current	–	50	70	90	μA	LOAD_SOLx	P_6.6.54
I_LS_PU	Diagnostic pull up current	–	55	75	95	μA	LOAD_SOLx	P_6.6.55
td_blank_00	Blank time 0	Bit_blank_x = 00, Guaranteed by scan		384		μs	LOAD_SOLx	P_6.6.56
td_blank_01	Blank time 1	Bit_blank_x = 01, Guaranteed by scan		768		μs	LOAD_SOLx	P_6.6.57
td_blank_10	Blank time 0	Bit_blank_x = 10, Guaranteed by scan		3.125		ms	LOAD_SOLx	P_6.6.58
td_blank_11	Blank time 1	Bit_blank_x = 11, Guaranteed by scan		6.25		ms	LOAD_SOLx	P_6.6.59
tflt_diagoff	Filter time	Guaranteed by scan	55	–	80	μs	LOAD_SOLx	P_6.6.60
	Minimum OFF time for correct diagnostic (Blank time 00)	Application note (esd cap < 6 nF, Bit_blank=00)	3.3	–		ms	LOAD_SOLx	
	Minimum OFF time for correct diagnostic (Blank time 01)	Application note (esd cap < 12 nF, Bit_blank = 01)	3.3	–		ms	LOAD_SOLx	
	Minimum OFF time for correct diagnostic (Blank time 10)	Application note (esd cap < 6 nF, Bit_blank = 10)	3.83	–		ms	LOAD_SOLx	

	Minimum OFF time for correct diagnostic (Blank time 11)	Application note (esd cap < 12 nF, Bit_blank = 11)	7.58	–		ms	LOAD_SOLx	
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Table 6-9 Valve Driver HS Off State Diagnostic Parameters

Symbol	Parameter	Test condition	Min	Typ	Max	Unit	Pin	Number
Off State Diagnostic – HS Configuration								
VTVBAT	Short to battery threshold voltage	HS mode, driver tristate, diagnostic on	2.7	2.9	3.1	V	LOAD_SOLx	P_6.6.65
VTOPEN	Open load threshold voltage	HS mode, driver tristate, diagnostic on	1.9	2.1	2.3	V	LOAD_SOLx	P_6.6.66
VOUTOPEN	Open load voltage	HS mode, driver tristate, diagnostic on	2.35	2.5	2.65	V	LOAD_SOLx	P_6.6.67
I_HS_THOPEN	Open load threshold current	–	30	60	100	μA	LOAD_SOLx	P_6.6.68
I_HS_PU	Diagnostic pull up current	–	55	70	95	μA	LOAD_SOLx	P_6.6.69
I_HS_PD	Diagnostic pull down current	–	60	70	100	μA	LOAD_SOLx	P_6.6.70
td_blank_00	Blank time 0	Bit_blank_x = 00, Guaranteed by scan		384		μs	LOAD_SOLx	P_6.6.71
td_blank_01	Blank time 1	Bit_blank_x = 01, Guaranteed by scan		768		μs	LOAD_SOLx	P_6.6.72
td_blank_10	Blank time 0	Bit_blank_x = 10, Guaranteed by scan		3.125		ms	LOAD_SOLx	P_6.6.73
td_blank_11	Blank time 1	Bit_blank_x = 11, Guaranteed by scan		6.25		ms	LOAD_SOLx	P_6.6.74
tflt_diagoff	Filter time	Guaranteed by scan	55	–	80	μs	LOAD_SOLx	P_6.6.75
	Minimum OFF time for correct diagnostic (Blank time 00)	Application note (esd cap < 6 nF, Bit_blank=00)	3.3	–		ms	LOAD_SOLx	
	Minimum OFF time for correct diagnostic (Blank time 01)	Application note (esd cap < 12 nF, Bit_blank = 01)	3.3	–		ms	LOAD_SOLx	
	Minimum OFF time for correct diagnostic (Blank time 10)	Application note (esd cap < 6 nF, Bit_blank = 10)	3.83	–		ms	LOAD_SOLx	
	Minimum OFF time for correct diagnostic (Blank time 11)	Application note (esd cap < 12 nF, Bit_blank = 11)	7.58	–		ms	LOAD_SOLx	

6.7 Valve driver electrical specifications (LOAD_SOL_5-14)

$4.75\text{ V} \leq \text{VDD} \leq 5.5\text{ V}$; $5.5\text{ V} \leq \text{VBAT} \leq 19\text{ V}$; $\text{VS_SOLx} \leq 19\text{ V}$; $\text{VBAT}-1\text{ V} \leq \text{VS_SOLx} \leq \text{VBAT}+1\text{ V}$;
 $-40\text{ }^{\circ}\text{C} \leq \text{Tj} \leq 175\text{ }^{\circ}\text{C}$ unless otherwise noted. All voltages refer to GND pin.

Table 6-10 Valve driver power stage electrical parameters

Symbol	Parameter	Test condition	Min	Typ	Max	Unit	Pin	Number
RDS-on_LS	Drain-source resistance	$\text{I}_{\text{load}} = 1\text{ A}$ $\text{Tj} = 150\text{ }^{\circ}\text{C}$	-	-	0.3	Ω	LOAD_SOLx	P_6.7.1
IOUT_LK_LS	Output leakage current	$\text{OUTx} = 13.5\text{ V}$	-	-	10	μA		P_6.7.2
VS/R_LS_SR00	Voltage rate00 slew	$\text{VB} = 13.5\text{ V}$ Load: $8\text{ }\Omega$, 10 nF – From 80% to 20% of OUTx	0.6	1	2	$\text{V}/\mu\text{s}$		P_6.7.3
VS/R_LS_SR01	Voltage rate01 slew	$\text{VB} = 13.5\text{ V}$ Load: $8\text{ }\Omega$, 10 nF – From 80% to 20% of OUTx	2	4	6	$\text{V}/\mu\text{s}$		P_6.7.5
VS/R_LS_SR10	Voltage rate10 slew	$\text{VB} = 13.5\text{ V}$ Load: $8\text{ }\Omega$, 10 nF – From 80% to 20% of VOUT	5	10	15	$\text{V}/\mu\text{s}$		P_6.7.7
VS/R_LS_SR11	Voltage rate11 slew	$\text{VB} = 13.5\text{ V}$ Load: $8\text{ }\Omega$, 10 nF – From 80% to 20% of OUTx	14	20	32	$\text{V}/\mu\text{s}$		P_6.7.9
LOAD_SOLx_ton00_LS	ton00_LS_mode	$\text{VB} = 13.5\text{ V}$ Load: $8\text{ }\Omega$, 10 nF – from INx or CS to 80% output	2.6	4.6	7.3	μs		P_6.7.11
LOAD_SOLx_ton01_LS	ton01_LS_mode	$\text{VB} = 13.5\text{ V}$ Load: $8\text{ }\Omega$, 10 nF	0.7	1.3	2.0	μs		P_6.7.12

		nF – from INx or CS to 80% output						
LOAD_SOLx_ton10_LS	ton10_LS_mode	VB=13.5V Load: 8 Ω , 10 nF – from INx or CS to 80% output	0.5	0.9	1.4	μ s		P_6.7.13
LOAD_SOLx_ton11_LS	ton11_LS_mode	VB=13.5V Load: 8 Ω , 10 nF – from INx or CS to 80% output	0.2	0.4	1	μ s		P_6.7.14
LOAD_SOLx_toff00_LS	toff00_LS_mode	VB=13.5V Load: 8 Ω , 10 nF – from INx or CS to 20% output	0.9	2.8	5.7	μ s		P_6.7.15
LOAD_SOLx_toff01_LS	toff01_LS_mode	VB=13.5V Load: 8 Ω , 10 nF – from INx or CS to 20% output	0.8	1.7	2.9	μ s		P_6.7.16
LOAD_SOLx_toff10_LS	toff10_LS_mode	VB=13.5V Load: 8 Ω , 10 nF – from INx or CS to 20% output	0.7	1.2	1.9	μ s		P_6.7.17
LOAD_SOLx_toff11_LS	toff11_LS_mode	VB=13.5V Load: 8 Ω , 10 nF – from INx or CS to 20% output	0.5	0.9	1.5	μ s		P_6.7.18
VCL_LS	Output clamping voltage	I_load = 0.6 A T = 130 $^{\circ}$ C	34	37	40	V	LOAD_SOLx	P_6.7.20
		I_load = 0.6A T = -40 $^{\circ}$ C and 25 $^{\circ}$ C	35	37	40	V		P_6.7.21
EclampSP_LS	Energy repetitive pulse	LS configuration						P_6.7.22
		I_load = 0.7 A, T _j = 150 $^{\circ}$ C,	-	-	5	mJ		P_6.7.23
		100 kpulses						P_6.7.24
EclampSP_LS	Single energy pulse	LS configuration						P_6.7.25
		I_load = 0.7 A, T _j = 150 $^{\circ}$ C	-	-	10	mJ		P_6.7.26

Table 6-11 Valve driver on state diagnostics parameters

Symbol	Parameter	Test condition	Min	Typ	Max	Unit	Pin	Number
On state diagnostic								
I_LS_ovc0	LS overcurrent	OVC_thres_x=0	3	–	–	A	LOAD_SOLx	P_6.7.27
I_LS_ovc1	LS overcurrent	OVC_thres_x=1	4.3	–	–	A	LOAD_SOLx	P_6.7.28
I_LS_ovc_ft	LS overcurrent filter time	Guaranteed by scan	3	–	5	μs	LOAD_SOLx	P_6.7.29
T_WARN	Temperature warning	–	175	–	185	°C	Tj_SOLx	P_6.7.30
T_SD	Temperature shut down	–	185	–	200	°C	Tj_SOLx	P_6.7.31
T_SD_deglitch	Digital deglitch filter time	Guaranteed by scan	7.5	10	12.5	μs	Tj_SOLx	P_6.7.32
T_WARN_deglitch	Digital deglitch filter time	Guaranteed by scan	7.5	10	12.5	μs	Tj_SOLx	P_6.7.33
Inx_ft_00	Inx filter time_00	Inx ft=00		0		μs	Inx	P_6.7.34
Inx_ft_01	Inx filter time_01	Inx ft=01		0.5		μs	Inx	P_6.7.35
Inx_ft_10	Inx filter time_10	Inx ft=10		1		μs	Inx	P_6.7.36
Inx_ft_11	Inx filter time_11	Inx ft=11		2		μs	Inx	P_6.7.37
Vclamp	LOADx clamp voltage	VBAT < 27V	34	37	40	V	LOAD_SOLx	P_6.7.38
Vclamp_dly	LOADx disable delay by clamp activation detection	Design info, not tested	–	–	2	μs	LOAD_SOLx	P_6.7.39
Vclamp_ft	LOADx clamp detection filter time	Guaranteed by scan	3.5	5	7	μs	LOAD_SOLx	P_6.7.40
Eclamp	Energy clamp single pulse	Iload = 1 A, Tj = 100 °C	–	–	25	mJ	LOAD_SOLx	P_6.7.41
Eclamp	Energy clamp single pulse	Iload = 1 A, Tj = 175 °C	–	–	18	mJ	LOAD_SOLx	P_6.7.42

Table 6-12 Valve driver off state diagnostics parameters

Symbol	Parameter	Test condition	Min	Typ	Max	Unit	Pin	Number
Off state diagnostic – LS configuration								

VTGND	Short to GND threshold voltage	LS mode, driver tristate, diagnostic on	1.9	2.1	2.3	V	LOAD_SOLx	P_6.7.43
VTOPEN	Open load threshold voltage	LS mode, driver tristate, diagnostic on	2.7	2.9	3.1	V	LOAD_SOLx	P_6.7.44
VOUTOPEN	Open load voltage	LS mode, driver tristate, diagnostic on	2.35	2.5	2.65	V	LOAD_SOLx	P_6.7.45
I_LS_THOPEN	Open load threshold current	–	40	60	90	μA	LOAD_SOLx	P_6.7.46
I_LS_PD	Diagnostic pull down current	–	50	70	90	μA	LOAD_SOLx	P_6.7.47
I_LS_PU	Diagnostic pull up current	–	55	75	95	μA	LOAD_SOLx	P_6.7.48
td_blank_00	Blank time 0	Bit_blank_x = 00, Guaranteed by scan		384		μs	LOAD_SOLx	P_6.7.49
td_blank_01	Blank time 1	Bit_blank_x = 01, Guaranteed by scan		768		μs	LOAD_SOLx	P_6.7.50
td_blank_10	Blank time 0	Bit_blank_x = 10, Guaranteed by scan		3.125		ms	LOAD_SOLx	P_6.7.51
td_blank_11	Blank time 1	Bit_blank_x = 11, Guaranteed by scan		6.25		ms	LOAD_SOLx	P_6.7.52
tflt_diagoff	Filter time	Guaranteed by scan	55	-	80	μs	LOAD_SOLx	P_6.7.53
	Minimum OFF time for correct diagnostic (Blank time 00)	Application note (esd cap < 6 nF, Bit_blank=00)	3.3	–		ms	LOAD_SOLx	
	Minimum OFF time for correct diagnostic (Blank time 01)	Application note (esd cap < 12 nF, Bit_blank = 01)	3.3	–		ms	LOAD_SOLx	
	Minimum OFF time for correct diagnostic (Blank time 10)	Application note (esd cap < 6 nF, Bit_blank = 10)	3.83	–		ms	LOAD_SOLx	
	Minimum OFF time for correct diagnostic (Blank time 11)	Application note (esd cap < 12 nF, Bit_blank = 11)	7.58	–		ms	LOAD_SOLx	

7 Safety features

The HE9314 has several integrated features to enable a reliable functional safety system implementation. A complete list of all monitoring and self-test functions is shown in the next table. The following sections detail these features.

Table 7-1 Safety monitors and self test summary

	Function	Primary monitor	Self test
Supply monitors	VBAT	OV&UV	POR
	VCP	UV	POR
	VDD	OV & UV	POR
Global functions internal monitors	V3V3AN	OV & UV	POR
	V3V3Dig	OV & UV	POR
	Trimming Data Error	CRC	None
	SPI	Timeout	None
	SPI	n * Timeout Latch	None
	Core Temperature	Over Temperature	None
	Oscillator	Diag Clk stuck-at	None
	Oscillator	Clock Mismatch + Main Clk stuck-at	None
Solenoid channel internal monitor	GND_SOLx	LOSS	POR & On Demand
	HS (On state)	OC	None
	LS (On State)	OC	None
	LS Clamp (Only in LS conf)	OV (dis. for VB > VBAT_OV_dis)	None
	STG, Off State (Only in LS conf)		On Demand Only
	Open Load, Off State		On Demand Only
	STB, Off State (Only in HS conf)		On Demand Only
	Driver Temperature	Over Temperature	None
	Driver Status	Command vs. Actual	On Demand Only

7.1 Power supply monitoring and diagnostics

7.1.1 VDD over and under voltage diagnostics

The VDD power supply input pin is continuously monitored for over and under voltage conditions by comparing the input voltage with an independent reference. In addition to the static voltage monitoring function, the HE9314 includes a self-test to verify the proper operation of the comparator and fault logic thus preventing latent faults from influencing device operation. The self-test is run automatically after a power on reset event. Upon detecting a fault, the corresponding fault bits are set as summarized below.

Table 7-2 VDD Fault Summary

Register	Bit Name
servflt1	VDD Under Voltage
servflt1	VDD Over Voltage
odtests 0	VDD UV Self Test
odtests 0	VDD OV Self Test

A VDD under voltage fault will disable all solenoid drivers and CP. The fault is clear on SPI read but solenoid operation must be re-enabled through the SPI.

In case of overvoltage on VDD, fault information is available on SPI, but no action is executed on the solenoid drivers. The fault is clear on SPI read.

7.1.2 VBAT supply over voltage

The VBAT power supply input pin is continuously monitored for an over voltage condition by comparing the input voltage with an independent reference. In addition to the static voltage monitoring function, the HE9314 includes a self-test to verify the proper operation of the comparator and fault logic thus preventing latent faults from influencing device operation. The self-test is run automatically after a power on reset event. Upon detecting a fault, the corresponding fault bits are set as summarized below.

Table 7-3 VBAT Fault Summary

Register	Bit Name
servflt1	VBAT over voltage
odtests 0	VBAT over voltage self-test
odtests 0	VBAT over voltage warning
servflt1	VBAT under voltage
odtests 0	VBAT under voltage self-test

A VBAT over voltage warning ,fault information is available on SPI, but no action is executed on the solenoid drivers. The fault is clear on SPI read.

A VBAT over voltage fault will disable all solenoid drivers and CP. The fault is clear on SPI read but solenoid operation must be re-enabled through the SPI.

A VBAT under voltage , fault information is available on SPI, but no action is executed on the solenoid drivers. The fault is clear on SPI read.

7.1.3 VCP supply under voltage

The VCP charge pump power supply is continuously monitored for an under voltage condition by

comparing the input voltage with an independent reference. Upon detecting a fault, the corresponding fault bit is set as summarized below.

Table 7-4 Charge pump fault summary

Register	Bit Name
servflt1	CP under voltage

A VCP under voltage fault will disable channel drivers 1-4. The fault is cleared on SPI read but solenoid operation must be re-enabled through the SPI.

7.1.4 Bandgap regulators, internal analog and digital supply monitoring

There are two internal bandgap regulators, main and auxiliary and two internal low voltage regulators, V3VA and V3VD. These regulators are monitored for over and under voltage conditions and a fault on any regulator regardless of the type of the result in a Power On Reset (POR). There are no flags dedicated to these fault conditions other than the POR flag. A POR event resets all registers and disables all solenoids. The POR flag is available as shown below.

Table 7-5 POR fault summary

Register	Bit Name
servflt1	POR Flag

In addition to the static voltage monitoring function, the HE9314 includes a self-test to verify the proper operation of all comparators and fault logic thus preventing latent faults from influencing the device operation. The self-test function verifies the voltage comparison and logic to ensure the entire path is verified.

7.1.5 GND loss detection

GND is continuously monitored for loss of ground connection. Ground loss comparators and fault logic are monitored for latent faults using a self-test function. The self-test is automatically performed upon a POR event. Should a ground loss or self-test fault be detected, the corresponding fault bits are set and are summarized below.

Each solenoid driver channel has a dedicated ground connection, GND_SOL0-10, PGND1-2. Should a fault be detected, the faulted channel will be disabled and fault flag set as shown below:

Table 7-6 Solenoid GND fault summary

Register	Bit Name
exceptions	Ground Loss

Each solenoid ground connection is continuously monitored for loss of connection. Each loss of ground monitoring circuit also contains a self-test function that must be requested by the user. Both `drv_diag_self_test` and `gndloss_self_test` are activated through `self_test_en` bit, but `gndloss` also performs `self_test` during POR.

Power supply monitoring and diagnostics electrical characteristics

4.75 V ≤ VDD ≤ 5.5 V; 5.5 V ≤ VBAT ≤ 19 V; -40 °C ≤ Tj ≤ 175 °C unless otherwise noted. All voltages refer to GND pin.

Table 7-7 Power supply monitoring electrical performance

Symbol	Parameter	Test condition	Min	Typ	Max	Unit	Pin	Number
VDD_UV	VDD under voltage detection	–	3.6	–	3.9	V	VDD	P_7.1.6.1
VDD_UV_hyst	VDD under voltage hysteresis	–	VDD_UV +0.1	–	VDD_U +0.5	V	VDD	P_7.1.6.2
VDD_UV_deglitch	Digital filter time	Guaranteed by scan	9	–	12	µs	VDD	P_7.1.6.3
VDD_OV	VDD over voltage detection	–	5.5	–	7	V	VDD	P_7.1.6.4
VDD_OV_hyst	VDD over voltage hysteresis	–	VDD_OV -0.1	–	VDD_O -0.5	V	VDD	P_7.1.6.5
VDD_OV_deglitch	Digital filter time	Guaranteed by scan	90	–	120	µs	VDD	P_7.1.6.6
VBAT_OV_th1	VBAT over voltage warning		27	28	29			P_7.1.6.7
VBAT_OV_th1_hyst	VBAT over voltage warning hysteresis		VBAT_OV_th1-1	-	VBAT_OV_th1-2			P_7.1.6.8
VBAT_OV_th1_deglitch	Digital filter time	Guaranteed by scan	90	–	120	µs	VBAT	P_7.1.6.9
VBAT_OV_dis ⁽¹⁾	VBAT over voltage detection	–	31	33	35	V	VBAT	P_7.1.6.10
VBAT_OV_hyst	VBAT over voltage hysteresis	–	VBAT_OV -1	-	VBAT_OV -3	V	VBAT	P_7.1.6.11
VBAT_OV_deglitch	Digital filter time	Guaranteed by scan	90	–	120	µs	VBAT	P_7.1.6.12
VBAT_UV	VBAT under voltage detection	–	5.8	6	6.2	V	VBAT	P_7.1.6.13
VBAT_UV_hyst	VBAT under voltage hysteresis	–	VBAT_UV +0.1	–	VBAT_UV +0.5	V	VBAT	P_7.1.6.14
VBAT_UV_deglitch	Digital filter time	Guaranteed by scan	–	10	–	µs	VBAT	P_7.1.6.15
CP_UV	Charge Pump under voltage detection	–	2.4	–	–	V	CP	P_7.1.6.16
CP_UV_deglitch	Charge Pump Under Voltage Filter time	Guaranteed by scan	9	–	12	µs	CP	P_7.1.6.17
V3V3A_uv	Internal Analog supply under voltage monitor	Design info, not tested	2.6	–	–	V	V3V3A	P_7.1.6.18
V3V3A_uv_hys	Internal Analog supply under voltage hysteresis	–	V3V3A_uv +0.1	–	V3V3A_uv +0.5	V	V3V3A	P_7.1.6.19
V3V3A_ov	Internal Analog supply over voltage monitor	Design info, not tested	3.6	–	4.6	V	V3V3A	P_7.1.6.20
V3V3A_ov_hys	Internal Analog supply over voltage hysteresis	–	V3V3A_ov -0.1	–	V3V3A_ov -0.5	V	V3V3A	P_7.1.6.21
V3V3D_uv	Internal Digital supply under voltage monitor	Design info, not tested	2.6	–	–	V	V3V3D	P_7.1.6.22
V3V3D_uv_hys	Internal Digital supply under voltage hysteresis	–	V3V3D_uv +0.1	–	V3V3D_uv +0.5	V	V3V3D	P_7.1.6.23
V3V3D_ov	Internal Digital supply over voltage monitor	Design info, not tested	3.6	–	4.6	V	V3V3D	P_7.1.6.24

						Safety features		
V3V3D_ov_hys	Internal Digital supply over voltage hysteresis	–	V3V3D_o v -0.1	–	V3V3D_o v -0.5	V	V3V3D	P_7.1.6.2 5
GND_SOLx_lost	Power ground lost detection	Design info, not tested	0.3	–	1	V	GND_S OLx	P_7.1.6.2 6
GND_SOLx_lost_deg litch	Power ground deglitch	Guaranteed by scan	35	–	50	µs	GND_S OLx	P_7.1.6.2 7

1. LS clamp circuitry automatically disabled, see [Section 6.5.3](#)

7.2 Main & reference oscillators

The HE9314 has two internal clock oscillators, main and reference. The main clock oscillator provides the timing and control for all device operating functions. The reference clock oscillator provides the monitoring reference to verify the main oscillator. The monitor is able to check if one of the two oscillators is stuck and if there is a strong frequency mismatch between them.

If a stuck fault on diagnostic oscillator is detected, the device keeps working but reports the corresponding fault through SPI Service Fault 2 register . Basic implementation of clock monitor is shown in the next figure.

Oscillator monitor is able to detect either stuck condition of one of the two clocks or frequency error of one clock respect the other.

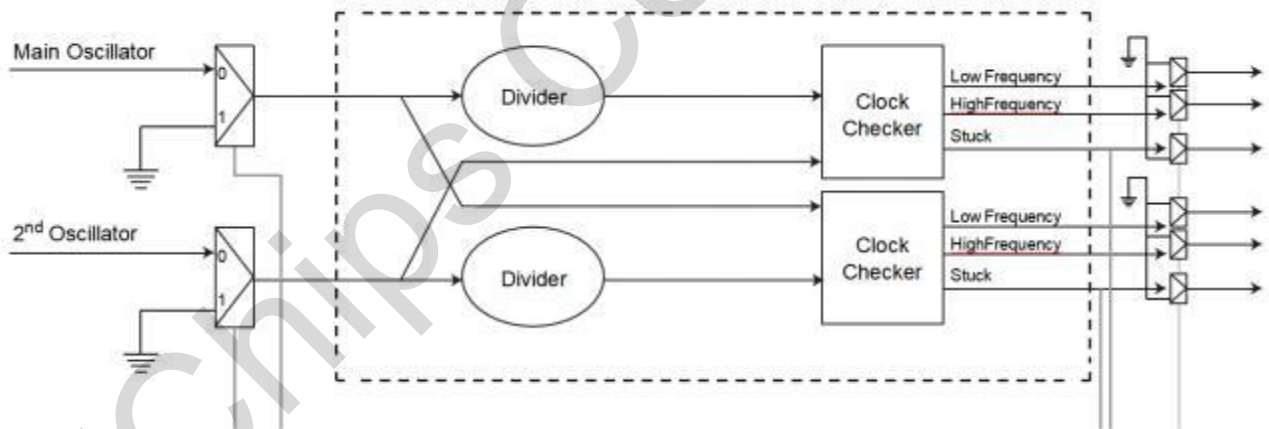


Figure 7-1 Oscillator monitor

7.2.1 Main and reference oscillators electrical characteristics

$4.75\text{ V} \leq \text{VDD} \leq 5.5\text{ V}$; $5.5\text{ V} \leq \text{VBAT} \leq 19\text{ V}$; $-40\text{ }^{\circ}\text{C} \leq \text{Tj} \leq 175\text{ }^{\circ}\text{C}$ unless otherwise noted. All voltages refer to GND pin.

Table 7-8 Main and reference oscillator safety parameters

Safety features

Symbol	Parameter	Test condition	Min	Typ	Max	Unit	Number
OSCMON_timeout	Timeout for stuck detection	Guaranteed by scan	–	–	60	μs	P_7.2.1.1
OSC_freq_error	Max tolerable delta between oscillators	Guaranteed by scan	-15	–	15	%	P_7.2.1.2

7.2.2 PWM check

The HE9314 can verify output driver status with respect to the PWM input signal. This function can only be performed when the channel is enabled and controlling load. Should a mismatch be detected, the PWM Check Fault, bit is set in the exceptions register.

7.2.3 Valve driver safety electrical characteristics

$5.5\text{ V} \leq \text{VBAT} \leq 19\text{ V}$; $5.5\text{ V} \leq \text{VSx} \leq 19\text{ V}$; $-40\text{ }^{\circ}\text{C} \leq \text{Tj} \leq 175\text{ }^{\circ}\text{C}$ unless otherwise noticed. All voltages refer to GND pin.

Table 7-9 Valve driver safety parameters

Symbol	Parameter	Test condition	Min	Typ	Max	Unit	Pin	Number
t_PWM_check	PWM check masking time	SR = 00 Guaranteed by scan	$13 \cdot 64 / f_{\text{OSCINT}}$	–	$14 \cdot 64 / f_{\text{OSCINT}}$	–	LOAD_SOLx	P_7.2.3.1
t_PWM_check	PWM check masking time	SR = 01 Guaranteed by scan	$9 \cdot 64 / f_{\text{OSCINT}}$	–	$10 \cdot 64 / f_{\text{OSCINT}}$	–	LOAD_SOLx	P_7.2.3.2
t_PWM_check	PWM check masking time	SR = 10/11 Guaranteed by scan	$5 \cdot 64 / f_{\text{OSCINT}}$	–	$6 \cdot 64 / f_{\text{OSCINT}}$	–	LOAD_SOLx	P_7.2.3.3
PWM_check_deglitch	PWM check fault deglitch time	Guaranteed by scan	0.2	–	1	μs	LOAD_SOLx	P_7.2.3.4

7.3 Core over temperature

The device internal temperature is monitored against overheating. If an over temperature event is detected, all solenoid driver outputs are disabled, their enable bit is cleared and their configuration registers are reset, a fault bit cleared on read is set in the SPI registers (Core Over Temperature bit) Once the over temperature condition is removed the fault information must be cleared to restart the outputs. Core over temperature is provided with a hysteresis in order to avoid power-on/power-off oscillations when temperature is around over temperature threshold.

7.3.1 Core over temperature electrical characteristics

$5.5\text{ V} \leq \text{VBAT} \leq 19\text{ V}$; $5.5\text{ V} \leq \text{VSx} \leq 19\text{ V}$; $-40\text{ }^{\circ}\text{C} \leq \text{Tj} \leq 175\text{ }^{\circ}\text{C}$ unless otherwise noticed. All voltages refer to GND pin.

Table 7-10 Core thermal monitoring electrical performance

Symbol	Parameter	Test condition	Min	Typ	Max	Unit	Number
CORE_ot	CORE over temperature shut down threshold	-	175	–	185	$^{\circ}\text{C}$	P_7.3.1.1
CORE_ot_hyst	CORE over temperature hysteresis	–	10	15	20	$^{\circ}\text{C}$	P_7.3.1.2
CORE_ot_filt	Digital deglitch	Guaranteed by scan	9	10	12	μs	P_7.3.1.3

8 Current Sense Amplifier

8.1 Overview

The current sense amplifier (CSA) is specially designed for current sensing in automotive applications. It is a unidirectional, single-supply amplifier for amplifying small differential voltages in a wide common mode voltage range. It supports the current measurement at four shunts.

The inputs (CSAx IN+ / CSAx IN-) are built as a transconductance stage. Therefore a series resistor (for filtering etc.) should not exceed 50 to keep the additional gain error below 1.5%.

8.2 Functional Block Diagram

$5.5\text{ V} \leq \text{VBAT} \leq 19\text{ V}$; $\text{CSAxIN+} = \text{VBAT}$, $4.75\text{ V} \leq \text{VDD} \leq 5.5\text{ V}$; $-40\text{ }^{\circ}\text{C} \leq \text{Tj} \leq 175\text{ }^{\circ}\text{C}$ unless otherwise noticed. All voltages refer to GND pin.

Table 8-1 X Current Sense Amplifier output electrical characteristics

Symbol	Parameter	Test condition	Min	Typ	Max	Unit	Pin	Number
INPUT								
CMRR	Common-mode rejection ratio, RTI	$V_{\text{IN+}} = 5.5\text{ to }19\text{ V}$, $V_{\text{SENSE}} = 10\text{ mV}$, $T_{\text{A}} = -40\text{ to }125\text{ }^{\circ}\text{C}$	84	100	-	dB	CSAxIN+, CSAxIN-	P_8.2.1
Vos	Offset Voltage, RTI	$V_{\text{OUT}} = 0.5\text{ V to VDD} - 0.5\text{ V}$, $T_{\text{A}} = -40\text{ to }125\text{ }^{\circ}\text{C}$	-	± 100	± 800	μV	CSAxIN+, CSAxIN-	P_8.2.2
dVos/dT	Offset drift, RTI	$T_{\text{A}} = -40\text{ to }125\text{ }^{\circ}\text{C}$	-	0.2	1	$\mu\text{V}/^{\circ}\text{C}$	CSAxIN+, CSAxIN-	P_8.2.3
I _{IB}	Input bias current	$V_{\text{SENSE}} = 0\text{ mV}$	-	500	-	μA	CSAxIN+, CSAxIN-	P_8.2.4
Output								
G	Gain	-	-	50	-	V/V	CSAxOUT	P_8.2.5
EG	Gain error	$V_{\text{OUT}} = 0.5\text{ V to VDD} - 0.5\text{ V}$, $T_{\text{A}} = -40\text{ to }125\text{ }^{\circ}\text{C}$	-	$\pm 0.1\%$	$\pm 1.5\%$	-	CSAxOUT	P_8.2.6
	Gain error vs temperature	$T_{\text{A}} = -40\text{ to }125\text{ }^{\circ}\text{C}$ Guaranteed by scan	-	1.5	20	ppm/ $^{\circ}\text{C}$	CSAxOUT	P_8.2.7
	Maximum capacitive load	No sustained oscillation		1	-	nF	CSAxOUT	P_8.2.8
Voltage Output								

Current Sense Amplifier

V _{SP}	Swing to V _{DD} power supply rail	R _L = 10 kΩ to GND, T _A = -40 to 125 °C	-	V _{DD} -0.2	V _{DD} -0.3	V	CSAxOUT	P_8.2.9
V _{SN}	Swing to GND	R _L = 10 kΩ to GND, T _A = -40 to 125 °C	-	(V _{GND}) +0.1	(V _{GND}) +0.15	V	CSAxOUT	P_8.2.10
Frequency Response								
BW	Bandwidth	-	-	200	-	kHz	CSAxIN+, CSAxIN-, CSAxOUT	P_8.2.11
SR	Slew Rate	-	-	0.2	-	V/μs	CSAxOUT	P_8.2.12

(1) RTI = referred-to-input.

(2) Offset voltage is obtained by linear extrapolation to V_{SENSE} = 0 V with V_{SENSE} = 10% to 90 % of full-scale-range.

(3) Swing specifications are tested with an overdriven input condition.

9 Queries and answer (Q&A) watchdog

9.1 Functional description

HE9314 and MCU exchanges queries and answers on a defined timing base.

An internal watchdog logic is implemented to inhibit load actuation such as to send reset signal.

A second switch off path, as shown in the figure below, is provided to allow the micro to force the switch off of the actuation channels.

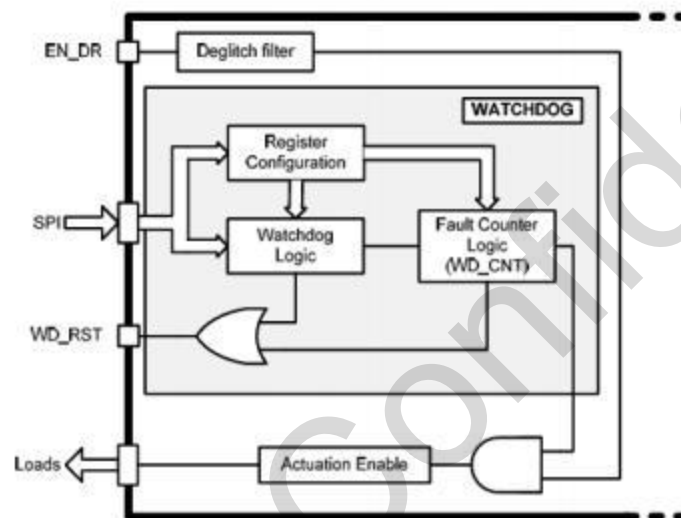


Figure 9-1 WD block diagram

9.2 Timing

Two modes of timing checks are provided:

Monodirectional: timing check based only on answers. MCU must send queries (without timing window check) and answers on a defined time window.

Bidirectional: timings are bidirectionally checked. HE9314 must receive queries on a defined time window. MCU must send answers on a defined time window.

In case time window are not respected an error is generated.

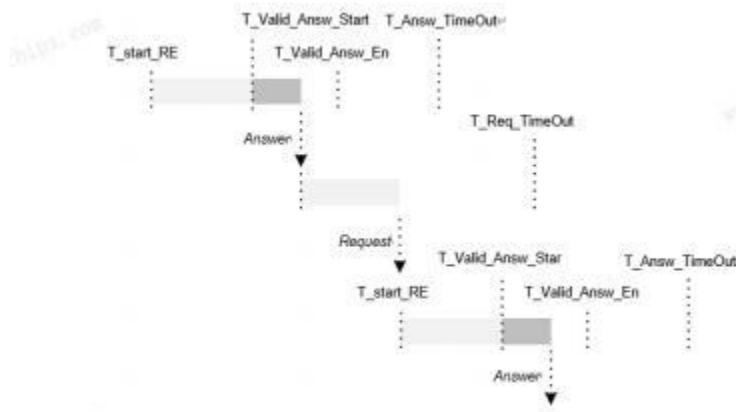


Figure 9-2 Monodirectional timing check evolution

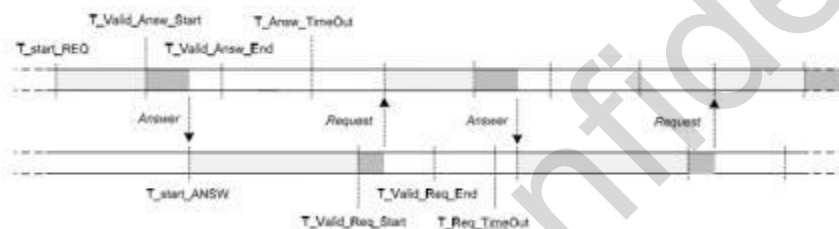


Figure 9-3 Bidirectional timing check evolution

Table 9-1 Timing

Timing parameter	description
T_start_REQ	Micro reads, through SPI register seed to elaborated
T_Valid_Answ_Start	Starting time interval for Valid answers
T_Valid_Answ_End	Ending time interval for Valid answers
T_Answ_TimeOut	Time out for answer
T_start_ANSW	Micro sends, through SPI register answer to the ASIC
T_Valid_Req_Start	Starting time interval for next following request
T_Valid_Req_End	Ending time interval for next following request
T_Req_TimeOut	Time out for request

Both the Request and the Answer must be sent on a predefined timing interval.

When the Micro finishes its boot procedures, it will send the first Seed sending request to the HE9314 with a read access on WDG2SEED register. In this moment all the timing counters will start and never stop.

In order to detect fast event, such as two consequent SPI frames, the time base is based on the WD frequency of about 250kHz; obtained from the ASIC clock period (16MHz). The obtained clock period is 4μs. The clock used for the timing windows is a divided version of that in order to obtain a timing resolution of WD_CLK equal to 64μs or 256μs depending on the WD_CLK_DIV settings.

Queries and answer (Q&A) watchdog

When the Micro sends the request of a new seed (with a write access on WDG2SEED register) to the HE9314 (T_start_REQ) the WD_ANSW_TMR timer starts to count.

Micro must send a valid answer inside the timing interval defined by the two SPI programmable parameters T_Valid_Answ_Start and T_Valid_Answ_End.

In case the Micro sends an answer before T_Valid_Answ_Start or after T_Valid_Answ_End an error will be generated.

In case the WD_TO_RST_EN is set

If no answer will arrive before T_Answ_TimeOut elapsed a WD_RST will be generated and the flag WD_RST_TO_Answ will be set.

When the Micro sends the answer to the HE9314 (T_start_ANSW) the WD_REQ_TMR timer starts to count.

Micro must send a new seed request inside the timing interval defined by the two SPI programmable parameters T_Valid_Req_Start and T_Valid_Req_End.

In case the Micro sends the request before T_Valid_Req_Start or after T_Valid_Req_End an error will be generated.

If no request will arrive before T_Req_TimeOut elapsed a WD_RST will be generated and the flag WD_RST_TO_Req will be set.

In case the WD_TO_RST_EN is not set:

The error event counter, WD_CNT, will be decreased and the device starts to wait again for the answer with the same timing procedure.

9.3 Watchdog states evolution

HE9314 starts the WD evolution state machine in IDLE mode where it's waiting for the first seed request from micro through SPI.

In this way the starting period is completely under the control of the Micro allowing to safely concluding boot procedure before starting the WD seed request/answer mechanism. During this period WD configuration registers can be programmed. The first seed request acts as a WD state machine start. After this event the WD will never stops and WD configuration registers becomes read only registers and cannot be changed. The only exception is about the T_Valid_Answ_Start and the T_Valid_Req_Start. In case one of these parameters is changed, the timing window restarts and WD_CNT will be decremented by a WD_cnt_bad_step number of steps. WD_CNT is a 4 bit counter used to collect good and bad events provided by the MCU.

A good event is defined in following conditions:

- A request coming in the correct timing window if a request is expected in the FSM;
- A correct answer coming in the correct timing window when expected.

A bad event is defined in following conditions:

- A wrong answer;
- A request in a wrong timing window (in bidirectional mode);
- An answer in a wrong timing window
- A timeout event;
- A request when an answer is expected;

Queries and answer (Q&A) watchdog

- An answer when a request is expected.

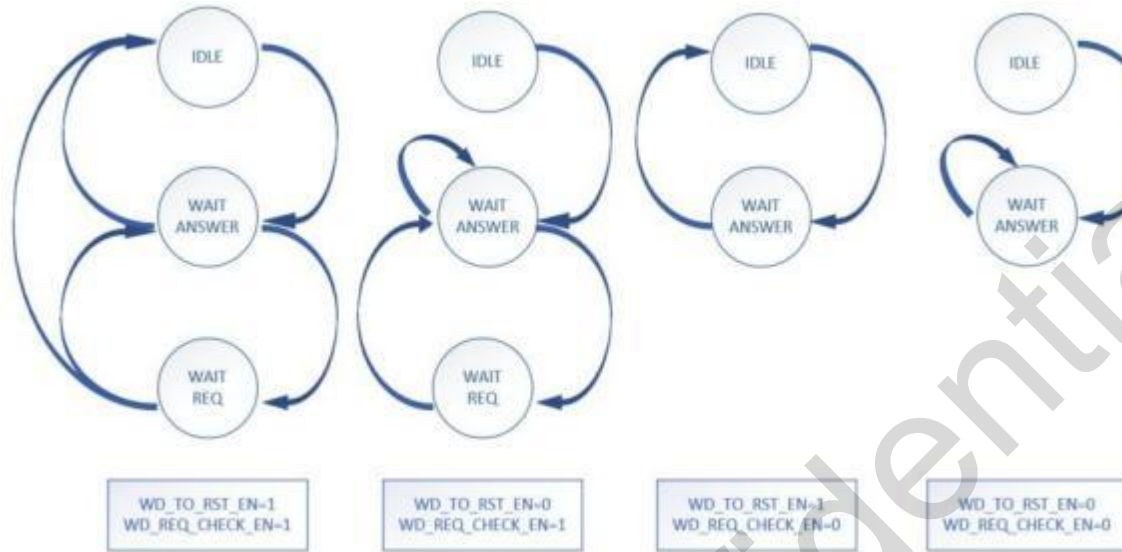


Figure 9-4 State evolution depending on WD_TO_RST_EN and WD_REQ_CHECK_EN

Note: Also in monodirectional mode the FSM is waiting a query after an answer in order to send a new seed to μ C but in this configuration the timing check on queries is not performed and WD_CNT is not decremented in case of request timing

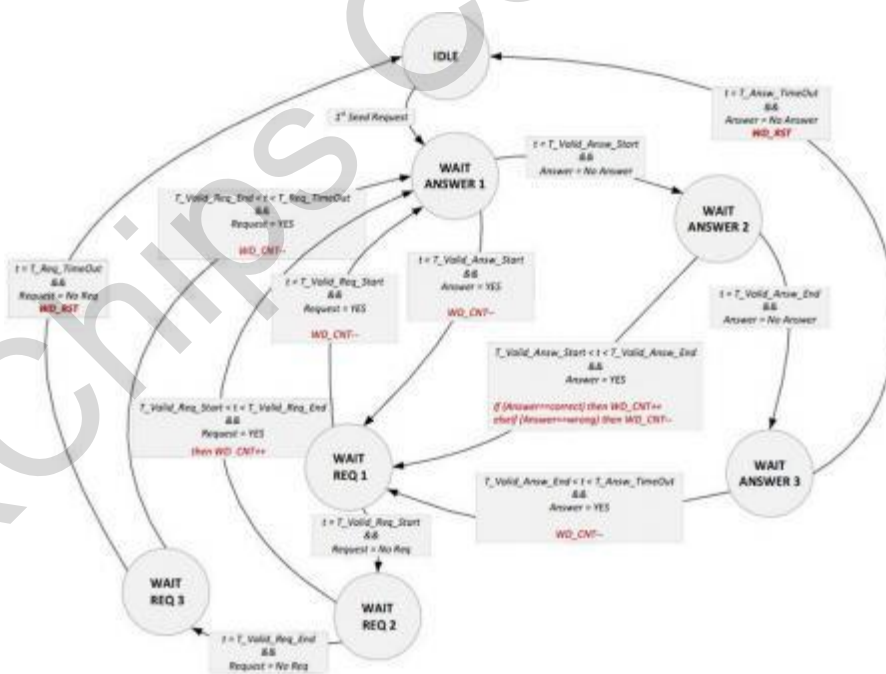


Figure 9-5 State evolution in case of WD_TO_RST_EN = 1

Depending on the value of the WD_CNT counter the HE9314 will stop the actuation, will send the WD_RST or will perform the actuation.

Queries and answer (Q&A) watchdog

The WD_CNT will be incremented, by a number of steps as defined through the SPI configurable parameter WD_cnt_good_step, when a correct answer is given in the right time interval.

In all the other cases, as defined in the WD state machine the WD_CNT will be decremented by a number of steps as defined through the SPI configurable parameter WD_cnt_bad_step. If WD_CNT reaches the value of zero two different behaviors are possible depending on the value of WD_RST_EN.

If WD_RST_EN is set to 1 then a WD_RST will be sent by the HE9314 and the flag WD_RST_Cnt will be set;

If WD_RST_EN is set to 0 then the WD_RST will not be sent by the HE9314 but the flag WD_RST_Cnt will not be set.

Two different thresholds are defined (both programmable through SPI): WD_th_low and WD_th_high. If WD_CNT value is lower than WD_th_low, but greater than zero, there will be no actuation such as any WD_RST.

If WD_CNT is greater than WD_th_low and lower than WD_th_high the load actuation is managed in hysteresis mode.

If actuation is ON it will be stopped only when WD_CNT becomes lower than WD_th_low, while if actuation was OFF it will be performed only when WD_CNT becomes equal to WD_th_high and only when WD_CNT exceeds this threshold loads are actuated.

In this way, actuation of loads can be performed only if the WD has been started and a certain number of good request/answer has been exchanged. HE9314

9.4 Unexpected error cases

Two cases of unexpected errors have been identified:

- If a request of a new seed arrives to the HE9314 before the previous answer received, the HE9314 will serve the new request, sending the old Seed decreasing the value of WD_CNT by the amount WD_cnt_bad_step.
- If an answer arrives to the HE9314 before a new request and after another answer, the HE9314 will ignore this answer but it will decrease the value of WD_CNT by the amount WD_cnt_bad_step.

9.5 Seed generation and answer check

Seeds are 8-bits wide word generated an LSFR pseudo-random algorithms. A new seed is generated and sent onto an SPI word, WD_Seed, each time a new seed request, T_start_REQ, is sent to the HE9314.

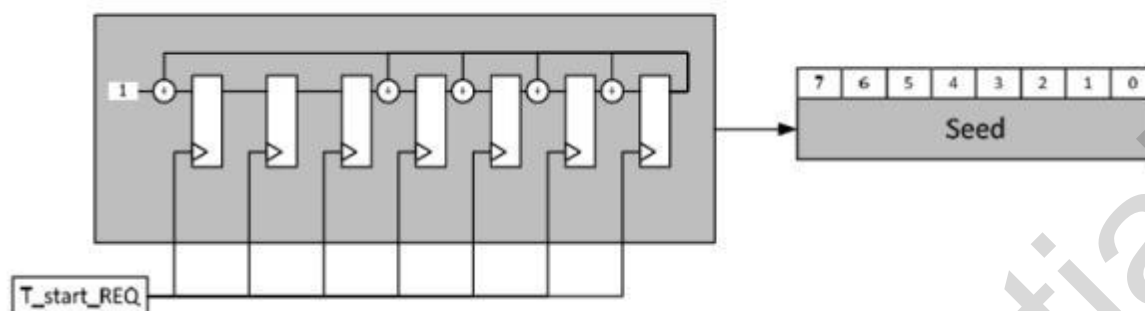


Figure 9-6 Seed generation algorithm block diagram

As described in figure, seed is generated by the LSFR algorithm. Each T_start_REQ the byte is one bit left shifted and the missing bit0, is obtained through the XOR operation between bit7 and bit6 of the previous byte. A new seed will be generated and stored onto the WD_Seed only in case of a correct answer received. In case of an error, the same seed will be available onto the WD_Seed until a correct answer will be received.

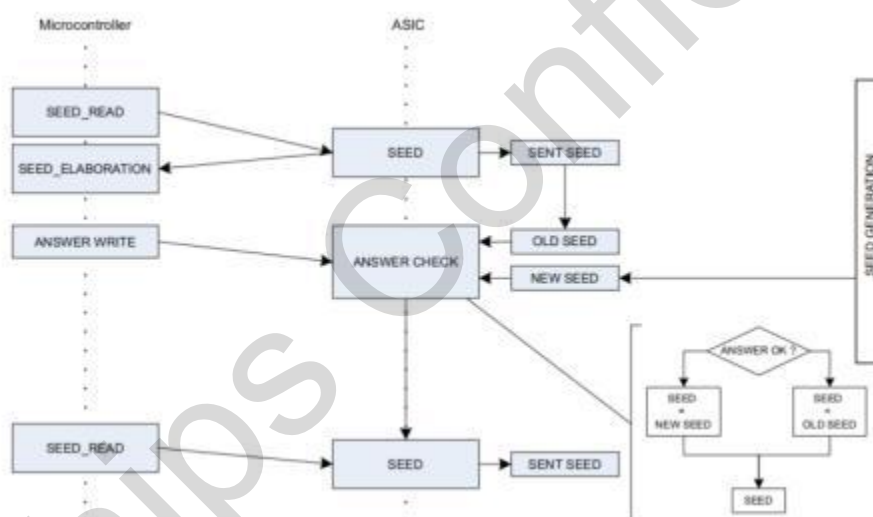


Figure 9-7 Seed selection and elaboration flow



Figure 9-8 Answer check generation algorithm block diagram

The answer is a 16-bits wide word checked against a 16-bits word composed by two bytes, Answer_Low and Answer_High, generated from the seed sent.

Answer_Low is the logical 2's complement of the seed while Answer_High is a replica of the seed sent.

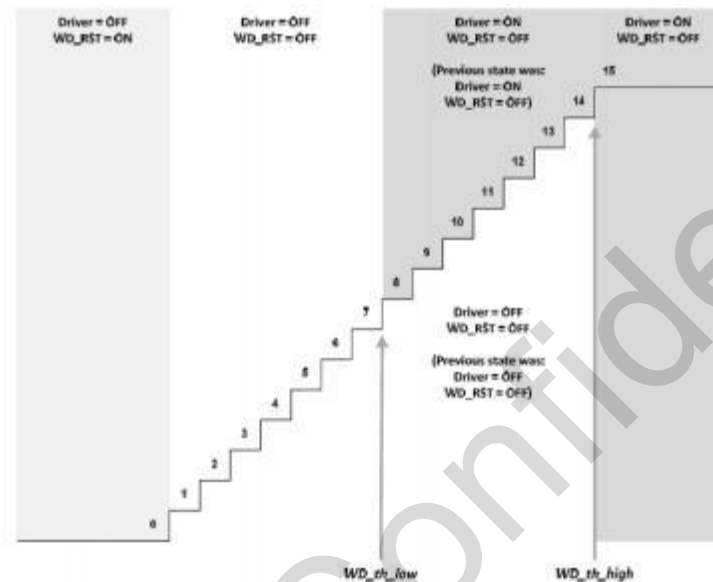


Figure 9-9 Driver actuation and WD_RTS versus WD_CNT value

All the status information are stored onto the WD_Status_reg, readable through SPI. This register will be cleaned as consequence of each read operation.

In case a WD_RST is sent to the micro, the HE9314 restarts the WD evolution state machine in IDLE mode waiting for the seed request from micro through SPI. WD configuration registers are preserved but can be modified by the micro before to start the WD mechanism. In the same way, also the status register, WD_Status_reg, is preserved and can be read through SPI.

9.6 WD SPI registers

WD_SEED

SPI parameter	Size (bits)	Description
SEED	8	Current value of the seed sent to the micro to be used for the answer elaboration.

WD_ANSW

SPI parameter	Size (bits)	Description
ANSWER_HIGH	8	Higher part of the answer obtained as described on par.
ANSWER_LOW	8	Lower part of the answer obtained as described on par.

WD_STATUS_REGISTER

SPI parameter	Size (bits)	Description
WD_RST_Event_Value (WD_RST_EVENT_VALUE)	4	Current value of the WD_RST event already sent.
WD_RST_Cnt (WD_RST_CNT)	1	Flag set if the WD_RST signal has been sent because error counter reached zero. 0 = Ok 1 = WD_RST because WD_CNT reached the value of zero
WD_RST_TO_Req (WD_RST_TO_REQ)	1	Flag set if the WD_RST signal has been sent because Request time out elapsed. 0 = Ok 1 = WD_RST because T_Req_Timeout elapsed
WD_RST_TO_AnsW (WD_RST_TO_ANSW)	1	Flag set if the WD_RST signal has been sent because Answer time out elapsed. 0 = Ok 1 = WD_RST because T_AnsW_Timeout elapsed
WD_Late_Req (WD_LATE_REQ)	1	Flag set if the last request has been sent too late related to the programmed timing parameters. 0 = Ok 1 = Request sent after T_Valid_Req_End time
WD_Early_Req (WD_EARLY_REQ)	1	Flag set if the last request has been sent too early related to the programmed timing parameters. 0 = Ok 1 = Request sent before T_Valid_Req_Start time
WD_Bad_AnsW (WD_BAD_ANSW)	1	Flag set if the last answer has been sent inside the right timing window (between T_Valid_AnsW_Start and T_Valid_AnsW_End time) but it is not the expected answer. 0 = Ok 1 = Wrong answer
WD_Late_AnsW (WD_LATE_ANSW)	1	Flag set if the last answer has been sent too late related to the programmed timing parameters. 0 = Ok 1 = Answer sent after T_Valid_AnsW_End time
WD_Early_AnsW (WD_EARLY_ANSW)	1	Flag set if the last answer has been sent too early related to the programmed timing parameters. 0 = Ok 1 = Answer sent before T_Valid_AnsW_Start time
WD_Cnt_Value (WD_CNT_VALUE)	4	Current value of the WD counter

WD_ANSW_TIMING

SPI parameter	Size (bits)	Description
WD_RST_EN	1	Enable for the WD_RST signal. 0 = WD_RST signal not sent 1 = WD_RST signal sent in case of failure
T_Valid_AnsW_End_Delta (VALID_ANSWER_DELTA)	6	End of the timing window inside which answers must be received. The value specified is an incremental time starting from T_Valid_AnsW_Start. Time = (T_Valid_AnsW_Start + T_Valid_AnsW_End_Delta)*WD_clk

Queries and answer (Q&A) watchdog

T_Valid_Answ_Start (VALID_ANSWER_START)	8	Start of the timing window inside which answers must be received. Absolute value. $\text{Time} = T_Valid_Answ_Start * WD_clk$
--	---	---

WD_REQ_TIMING:

SPI parameter	Size (bits)	Description
WD_REQ_CHECK_EN (REQ_CHECK_ENABLE)	1	Enable for the Request timing checking. 0 = Request timing check not performed 1 = Request timing check performed
T_Valid_Req_End_Delta (VALID_REQUEST_DELTA)	6	End of the timing window inside which requests must be received. The value specified is an incremental time starting from T_Valid_Req_Start. $\text{Time} = (T_Valid_Req_Start + T_Valid_Req_End_Delta) * WD_clk$
T_Valid_Req_Start (VALID_REQUEST_START)	8	Start of the timing window inside which requests must be received. Absolute value. $\text{Time} = T_Valid_Req_Start * WD_clk$

WD_COUNTER_SETUP

SPI parameter	Size (bits)	Description
WD_TO_RST_EN (TO_RESET_ENABLE)	1	Enable for the RST after Timeout generation. 0 = RST not generated after aTO event 1 = RST generated after aTO event
WD_CLK_DIV	1	Frequency Clock division setup. 0 = clock not divided (64 μ s) 1 = clock divided by (256 μ s)
WD_Th_Low	4	Threshold level to inhibit the actuation. If WD_CNT is lower than the threshold. no actuation will be performed.
WD_Th_High	4	Threshold level to start the actuation. If WD_CNT is lower than the threshold actuation will be performed depending on the previous state as shown in Figure 9-7.
WD_cnt_good_step (#_OF_GOOD_STEPS))	3	Number of incremental steps for WD_CNT as consequence of a good answer.
WD_cnt_bad_step (#_OF_BAD_STEPS)	3	Number of decremental steps for WD_CNT as consequence of an error.

WD_CONFIG

SPI parameter	Size (bits)	Description
T_VALID_ANSW_TO_DELTA (ANSWER_TIME_OUT_DELTA)	6	End of the period for answers acceptance. Once reached the WD_RST signal will be sent independently on the Error status (WD_CNT). The value specified is an incremental time starting from T_Valid_Answ_Start and T_Valid_Answ_End_Delta. $\text{Time} = (T_Valid_Answ_Start + T_Valid_Answ_End_Delta + T_Answ_TimeOut_Delta) * WD_clk$
T_VALID_REQ_TO_DELTA (REQUEST_TIME_OUT_DELTA)	6	End of the period for requests acceptance. Once reached the WD_RST signal will be sent independently on the Error status (WD_CNT). The value specified is an incremental time starting from T_Valid_Req_Start and T_Valid_Req_End_Delta. $\text{Time} = (T_Valid_Req_Start + T_Valid_Req_End_Delta + T_Req_TimeOut_Delta) * WD_clk$
wd_enable	1	Enable Watchdog 0=disable 1=enable
Disable uc reset	1	inhibit fault_low from WDA 0=disable (not inhibit) 1=enable (inhibit)
disable driver inhibit	1	inhibit driver from WDA 0=disable (not inhibit) 1=enable (inhibit)

Note:

When a read operation will be performed a new seed will be stored in the SPI shift register and be treated as a seed request, ready to be read by μ C by a read operation on the register.

A write operation (μ C sends the answer) on WD_ANSW register will be treated as a Answer Event

SPI parameters are programmable only before the WD mechanism starts. After the first seed request, these parameters can be only read.

A WD_RST event does not clear programmed parameters; default values are applied only as consequence of a POR

SPI parameters are programmable only before the WD mechanism starts. After the first seed request, these parameters can be only read.

Status register stores the information related to the last request and answer received so it is updated each time an answer and request is received.

Except the WD_CNT_VALUE and WD_RST_EVENT_VALUE fields, this register is automatically cleared once read.

10 SPI

The SPI interface is used to configure the device, control the outputs and read diagnostic and status registers.

10.1 SPI protocol

The SPI protocol is defined by 32-bit frames with 5 bits of CRC (Cyclic Redundancy Check) for transmit and receive. SDI frame consists of an 8-bit address, 16-bit data, 5-bit CRC field and a one-bit frame counter. SDO frame consists of SPI error and IC status information, address feedback, 16-bit data and 5-bit CRC field. SPI communication uses various safety mechanisms to ensure high safety level coverage as defined in ISO26262 table D.8:

- 5-bit CRC check (Hamming distance of 3 over 26 bits data)
- Frame Counter and Address Feedback
- Timeout Monitoring
- Short and Long Frame Length

The SPI frame format is shown in [Figure 10-1](#).

Table 10-1 SPI Frame Format

BIT	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
SDI	RW	Page			Index				0		cnt		Data[n]																CRC			
SDO	E	IERR		PageFdbk		IndexFdbk				Data[n-1]																CRC						

The protocol implements Out of Frame response meaning the SDO output frame is related to the previous transmitted SDO frame as shown in [Figure 10-1](#). The First SPI transfer after power on reset will have a fixed SDO response consisting in the device ID.

- *Note: Set the frm_cnt_dis bit to 1, and the Frame Counter will be disabled.*

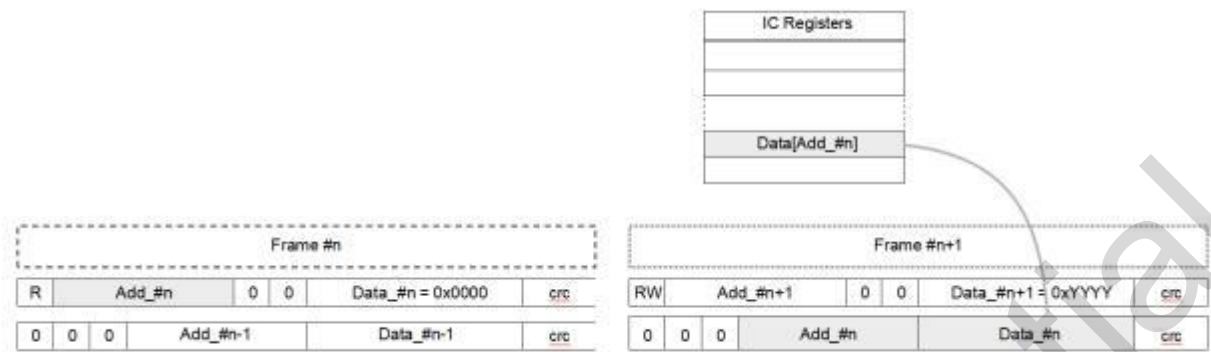


Figure 10-1 Out of frame response

Upon power on reset, the first SPI Frame will set all SPI fault bits and address bits to “0” and the data field will contain the device ID as shown in Figure 10-2.

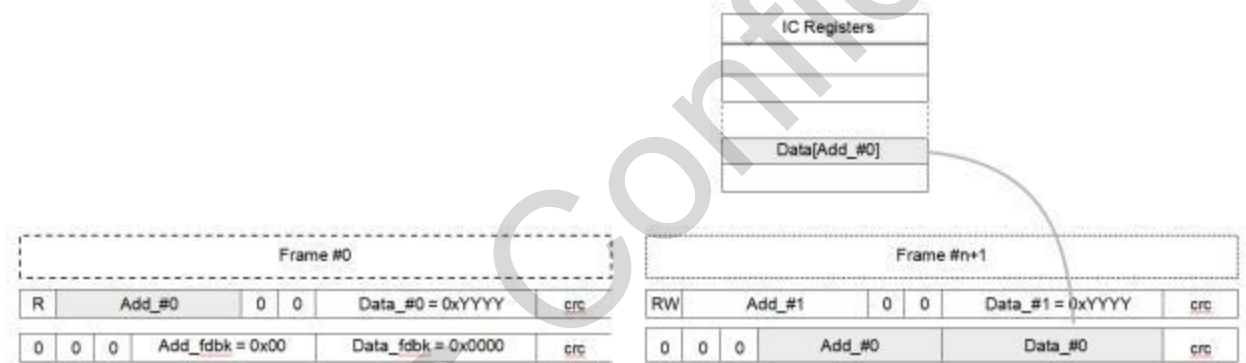


Figure 10-2 SPI frame after power on reset

Upon reading a register with Clear on Read fields, the related bits will be cleared only when data has been completely moved on the next SPI access as shown in Figure 10-3.



Figure 10-3 Clear on Read SPI sequence

Upon register write, data is updated after the frame is completely received and no error has been detected (CRC, frame sequence, frame length). On the next SPI access, SDO data field will contain updated data content from the written register as shown in Figure 10-4.

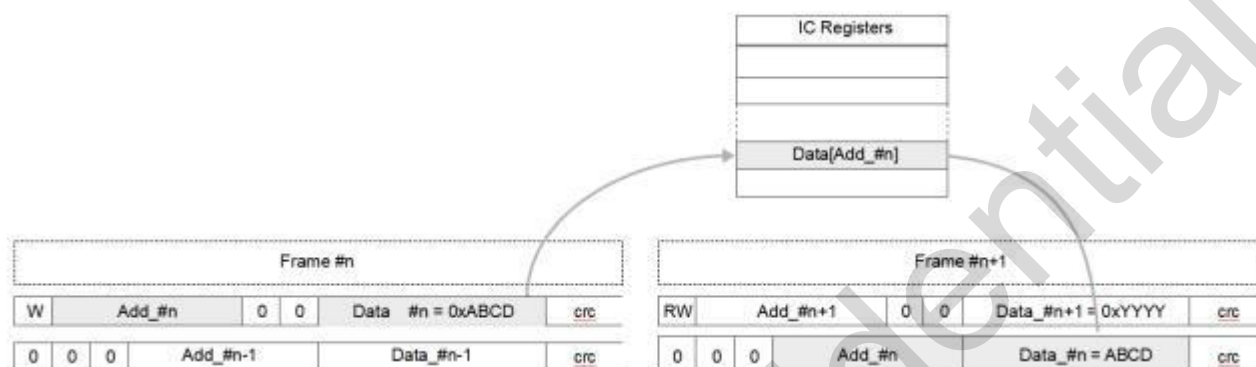


Figure 10-4 SPI write sequence

The SPI interface is supplied by VDDIO pin and consists of the following pins and definitions:

•Input pins:

–CS: Chip Select Active Low

–SCLK: SPI Clock

–SDI: Master Output Slave Input. Communication path from MCU to the device

•Output pins:

–SDO: Master Input Slave Output. Communication path from the device to the MCU

10.2 CRC

Received and transmitted data are verified using a 5-bit CRC field. The CRC is calculated from D[31] of the frame to D[5] excluding frame counter bit, D[21]. The CRC field is checked using 5° degree polynomial and is implemented per the following equation:

$$g(x) = x^5 + x^2 + 1$$

If a CRC error is detected, the received frame is ignored and the fault is indicated in the next SPI frame.

Bit 21 must be removed and bit 4 to bit 0 must be removed for CRC calculation.

10.3 SPI errors management

If a communication fault is detected, the SPI error bit (SDO[31]: E bit) is set to "1" and the frame is

ignored. The SPI error bit always refers to the previous SPI frame. The next figure shows the SPI frame sequence for faulted communications.

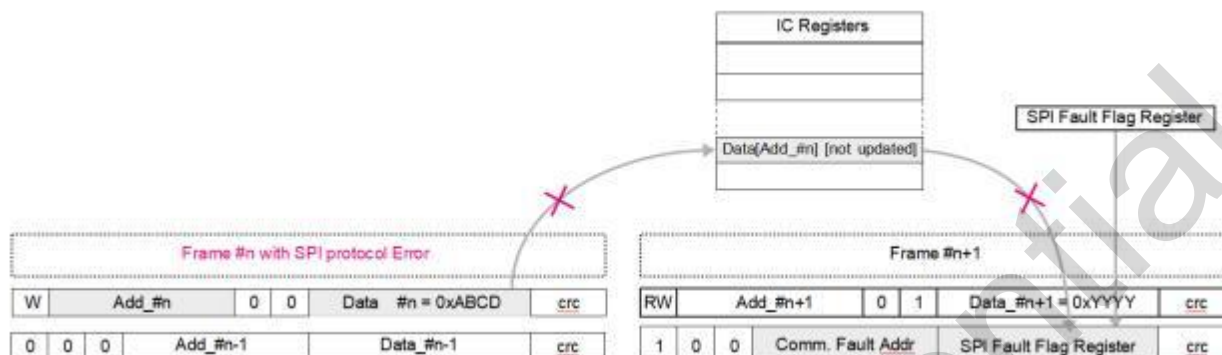


Figure 10-5 SPI frame with error on previous SPI frame

10.4 SPI timeout monitoring function

An internal monitor detects a failure if the period between communications exceeds the SPI Timeout period. This monitoring function detects loss of the communications between the micro controller and the device.

If no valid frames are received before the Timeout Configuration period elapses, the SPI Timeout flag is set and the FAULTn pin is set low if the corresponding fault mask bit is not set.

The Timeout Configuration value is programmable through a 3-bit SPI field (TIMEOUTSPICFG register) with a default value of 2ms. The timer is cleared when a valid SPI frame is received.

Table 10-2 SPI time out values

Dec	Bin	SPI Time Out 63 x 8 x 2N [μs]
0	000	504
1	001	1008
2	010	2016
3	011	4032
4	100	8064
5	101	16128
6	110	32256
7	111	64512

Reset latch function can reset the device to IDLE state if a certain number of timeout events occur. Enable of this function (TIMEOUTSPICFG register) and the number of timeout events (TIMEOUTSPICFG register) is programmed through SPI.

Table 10-3 Reset latch enable state

En_Res_Latch[1] (Reset Latch enable)	Dis_Res_Latch[0] (Reset latch disable)	Reset Latch
0	0	Disabled
0	1	Disabled
1	0	Enabled
1	1	Disabled

Table 10-4 Time out reset latch counter values

Dec	Bin	Time Out Reset Latch Counter [2N+1]
0	00	2
1	01	4
2	10	8
3	11	16

10.5 SPI channel exception field

SDO frame has 2 bits (IERR) reserved for solenoid channel exceptions. SDO[29] is the OR of all the exceptions related to Channel 1-7, while SDO[30] is the OR of all the exceptions related to Channel 8-14.

$\text{SDO}[29] = (\text{Ch1-7.Exceptionx} \mid \text{OTW} \mid \text{OTD})$

$\text{SDO}[30] = (\text{Ch8-14.Exceptionx} \mid \text{OTW} \mid \text{OTD})$

In this way, it is always possible to detect a failure happening on solenoid channel during every SPI communication. Details and clearing of the channel exception can be retrieved by reading the dedicated SPI registers.

10.6 SPI electrical specification

$4.75\text{ V} \leq \text{VDD} \leq 5.5\text{ V}$; $5.5\text{ V} \leq \text{VBAT} \leq 19\text{ V}$; $-40\text{ }^{\circ}\text{C} \leq \text{Tj} \leq 175\text{ }^{\circ}\text{C}$ unless otherwise noted. All voltages refer to GND pin.

Table 10-5 SPI electrical parameters

Symbol	Parameter	Test condition	Min	Typ	Max	Unit	Pin	Number
SPI_low	Logic input low detection	—	—	—	0.75	V	CS, SDI, SCLK	P_10.6.1
SPI_high	Logic input high detection	—	1.75	—	—	V	CS, SDI, SCLK	P_10.6.2

SPI

SPI_hyst	Input hysteresis voltage	–	0.1	–	–	V	CS, SDI, SCLK	P_10.6.3
SPI_pd	Internal pull down resistance	VIN < 2.5 V	20	40	80	kΩ	SDI, SCLK	P_10.6.4
SPI_pu	Internal pull up resistance	VIN < 2.5 V	40	90	180	kΩ	CS	P_10.6.5
SDO_ileak	Leakage current in tri-state condition	–	–	–	10	μA	SDO	P_10.6.6
SDO_low	Logic level low	Isink = 1 mA	–	–	0.4	V	SDO	P_10.6.7
SDO_high	Logic level high	Isource = 1 mA	VDDIO-0.4	–	–	V	SDO	P_10.6.8
SDO_trise	SDO rise time	Cload = 200 pF	–	–	50	ns	SDO	P_10.6.9
SDO_tfall	SDO fall time	Cload = 200 pF	–	–	50	ns	SDO	P_10.6.10
fclk	Transfer Frequency	Design info 50% cycle, duty not tested ⁽¹⁾	–	–	8	MHz	SCLK, SDI, SDO	P_10.6.11
tpcld+tsdo_trans	Propagation delay	SCLK to data at SDO is valid Cload_max = 200 pF including parasitic	–	–	100	ns	SCLK, SDO	P_10.6.12
tcsdv	NCS=LOW to data at SDO active	Cload_max = 200 pF including parasitic	–	–	200	ns	CS, SDO	P_10.6.13
tpchdz	NCS L/H to SDO at high impedance	Cload_max = 200 pF including parasitic	–	–	200	ns	CS, SDO	P_10.6.14
tsclch	SCLK before NCS low	Design info, not tested	50	–	–	ns	CS, SCLK	
thclcl	SCLK change L/H after NCS=LOW	Design info, not tested	130	–	–	ns	CS, SCLK	
tsclcl	SCLK before low NCS high	Design info, not tested	50	–	–	ns	CS, SCLK	
thclch	SCLK after high NCS	Design info, not tested	50	–	–	ns	CS, SCLK	

tsclhd	SDI input setup time	Design info, not tested	20	–	–	ns	SDI	
thclhd	SDI input hold time	Design info, not tested	–	–	20	ns	SDI	
tonNCS	NCS min. high time	Design info, not tested	650	–	–	ns	CS	
tclh	Minimum Time SCLK=HIGH	Design info, not tested	62.5	–	–	ns	SCLK	
tcll	Minimum Time SCLK=LOW	Design info, not tested	62.5	–	–	ns	SCLK	
Cin	Input pin capacitance	Design info, not tested	–	–	30	pF	SDI, CS, SCLK	
Cout_hiz	SDO output pin capacitance in tri-state	Design info, not tested	–	–	50	pF	SDO	

1. SPI max frequency may be less depending on the total capacitive load and MCU timing requirements

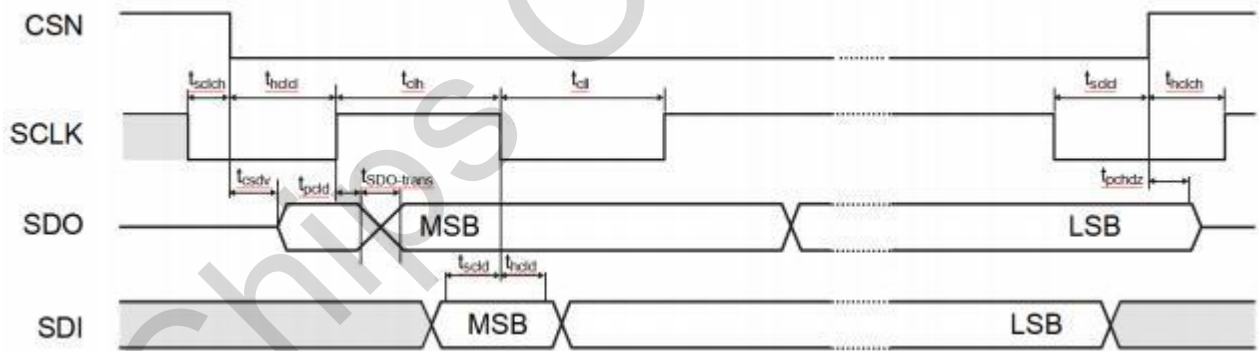


Figure 10-6 SPI timings, thresholds 20% to 80% and vice versa

10.7 SPI registers map table

SPI frame has an 8-bit address field and 16-bit data field, then the available address space consists of up to 256 different registers. Taking into account the logical and physical partition of the functions to be configured and monitored by SPI interface, the address space has been divided in pages of 32 addresses each.

The table below shows the SPI page vs. macro function distribution.

RESERVED:

ADDRESS	Name	TYPE	Notes
00000000	RESERVED	R/W	

Table 10-6 RESERVED

Data field	Description	Reset value	Reset event
Bit 15:0	RESERVED	0	

solenoid channel enable:

ADDRESS	Name	TYPE	Notes
00000001	Solenoid channel enable	R/W	

Table 10-7 solenoid channel enable

Data field	Description	Reset value	Reset event
Bit 15:14	NOT USED	0	POR/RESETn
Bit 13	driver_enable_13 0=disable 1=enable	0	EN_DR POR RESETn CLAMP_ON OVC T_SD CP_UV (only0-3) GND_LOSS VBAT_OV2 VDD_UV WDA_RESET Core_OT EN_DR SPI_Timeout Latch
Bit 12	driver_enable_12 0=disable 1=enable	0	
Bit 11	driver_enable_11 0=disable 1=enable	0	
Bit 10	driver_enable_10 0=disable 1=enable	0	
Bit 9	driver_enable_9 0=disable 1=enable	0	
Bit 8	driver_enable_8 0=disable 1=enable	0	
Bit 7	driver_enable_7 0=disable 1=enable	0	
Bit 6	driver_enable_6 0=disable 1=enable	0	
Bit 5	driver_enable_5 0=disable 1=enable	0	
Bit 4	driver_enable_4 0=disable 1=enable	0	
Bit 3	driver_enable_3 0=disable 1=enable	0	
Bit 2	driver_enable_2 0=disable	0	

	1=enable		
Bit 1	driver_enable_1	0	
	0=disable 1=enable		
Bit 0	driver_enable_0	0	
	0=disable 1=enable		

solenoid channel command:

ADDRESS	Name	TYPE	Notes
00000010	Driver control command	R/W	

Table 10-8 solenoid channel command

Data field	Description	Reset value	Reset event
Bit 15:14	NOT USED	0	POR/RESETn
Bit 13	driver_spi_on_13 0 = tristate 1 = active	0	EN_DR POR RESETn CLAMP_ON OVC T_SD CP_UV (only0-3) GND_LOSS VBAT_OV2 VDD_UV WDA_RESET Core_OT EN_DR SPI_Timeout Latch
Bit 12	driver_spi_on_12 0 = tristate 1 = active	0	
Bit 11	driver_spi_on_11 0 = tristate 1 = active	0	
Bit 10	driver_spi_on_10 0 = tristate 1 = active	0	
Bit 9	driver_spi_on_9 0 = tristate 1 = active	0	
Bit 8	driver_spi_on_8 0 = tristate 1 = active	0	
Bit 7	driver_spi_on_7 0 = tristate 1 = active	0	
Bit 6	driver_spi_on_6 0 = tristate 1 = active	0	
Bit 5	driver_spi_on_5 0 = tristate 1 = active	0	
Bit 4	driver_spi_on_4 0 = tristate 1 = active	0	
Bit 3	driver_spi_on_3 0 = tristate 1 = active	0	
Bit 2	driver_spi_on_2 0 = tristate 1 = active	0	
Bit 1	driver_spi_on_1 0 = tristate 1 = active	0	
Bit 0	driver_spi_on_0 0 = tristate	0	

	1 = active		
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solenoid channel command configuration:

ADDRESS	Name	TYPE	Notes
00000011	solenoid channel command configuration	R/W	

Table 10-9 solenoid channel command configuration

Data field	Description	Reset value	Reset event
Bit 15:10	NOT USED	0	POR/RESETn/SPI Time out Latch/ WDA_RESET/Core_OT
Bit 9	spi_ctl_sel_09 0= spi mode 1= in 10	0	POR/RESETn/SPI Time out Latch/ WDA_RESET/Core_OT
Bit 8	spi_ctl_sel_08 0= spi mode 1= in 9	0	POR/RESETn/SPI Time out Latch/ WDA_RESET/Core_OT
Bit 7	spi_ctl_sel_07 0= spi mode 1= in 8	0	POR/RESETn/SPI Time out Latch/ WDA_RESET/Core_OT
Bit 6	spi_ctl_sel_06 0= spi mode 1= in 7	0	POR/RESETn/SPI Time out Latch/ WDA_RESET/Core_OT
Bit 5	spi_ctl_sel_05 0= spi mode 1= in 6	0	POR/RESETn/SPI Time out Latch/ WDA_RESET/Core_OT
Bit 4	spi_ctl_sel_04 0= spi mode 1= in 5	0	POR/RESETn/SPI Time out Latch/ WDA_RESET/Core_OT
Bit 3	spi_ctl_sel_03 0= spi mode 1= in 4	0	POR/RESETn/SPI Time out Latch/ WDA_RESET/Core_OT
Bit 2	spi_ctl_sel_02 0= spi mode 1= in 3	0	POR/RESETn/SPI Time out Latch/ WDA_RESET/Core_OT
Bit 1	spi_ctl_sel_01 0= spi mode 1= in 2	0	POR/RESETn/SPI Time out Latch/ WDA_RESET/Core_OT
Bit 0	spi_ctl_sel_00 0= spi mode 1= in 1	0	POR/RESETn/SPI Time out Latch/ WDA_RESET/Core_OT

solenoid channel mode configuration

ADDRESS	Name	TYPE	Notes
00000100	solenoid channel mode configuration	R/W	

Table 10-10 solenoid channel mode configuration

Data field	Description	Reset value	Reset event
Bit 15:14	NOT USED	0	POR/RESETn/SPI Time out Latch/ WDA_RESET/Core_OT
Bit 13	full_on_enable_13 1=full on 0=full off	0	POR/RESETn/SPI Time out Latch/ WDA_RESET/Core_OT
Bit 12	full_on_enable_12 1=full on 0=full off	0	POR/RESETn/SPI Time out Latch/ WDA_RESET/Core_OT
Bit 11	full_on_enable_11 1=full on 0=full off	0	POR/RESETn/SPI Time out Latch/ WDA_RESET/Core_OT
Bit 10	full_on_enable_10 1=full on 0=full off	0	POR/RESETn/SPI Time out Latch/ WDA_RESET/Core_OT
Bit 9	full_on_enable_09 1=full on 0=full off	0	POR/RESETn/SPI Time out Latch/ WDA_RESET/Core_OT
Bit 8	full_on_enable_08 1=full on 0=full off	0	POR/RESETn/SPI Time out Latch/ WDA_RESET/Core_OT
Bit 7	full_on_enable_07 1=full on 0=full off	0	POR/RESETn/SPI Time out Latch/ WDA_RESET/Core_OT
Bit 6	full_on_enable_06 1=full on 0=full off	0	POR/RESETn/SPI Time out Latch/ WDA_RESET/Core_OT
Bit 5	full_on_enable_05 1=full on 0=full off	0	POR/RESETn/SPI Time out Latch/ WDA_RESET/Core_OT
Bit 4	full_on_enable_04 1=full on 0=full off	0	POR/RESETn/SPI Time out Latch/ WDA_RESET/Core_OT
Bit 3	full_on_enable_03 1=full on 0=full off	0	POR/RESETn/SPI Time out Latch/ WDA_RESET/Core_OT
Bit 2	full_on_enable_02 1=full on 0=full off	0	POR/RESETn/SPI Time out Latch/ WDA_RESET/Core_OT
Bit 1	full_on_enable_01 1=full on 0=full off	0	POR/RESETn/SPI Time out Latch/ WDA_RESET/Core_OT
Bit 0	full_on_enable_00 1=full on 0=full off	0	POR/RESETn/SPI Time out Latch/ WDA_RESET/Core_OT

Note: The priority of full_on mode higher PWM_mode

solenoid channel parallel mode configuration:

ADDRESS	Name	TYPE	Notes
00000101	solenoid channel parallel mode configuration	R/W	

Table 10-11 solenoid channel parallel mode configuration

Data field	Description	Reset value	Reset event
Bit 15	parallel_mode_sel_13_14 0=disable 1=enable	0	POR/RESETn/SPI Time out Latch/ WDA_RESET/Core_OT
Bit 14	parallel_mode_sel_11_12 0=disable 1=enable	0	POR/RESETn/SPI Time out Latch/ WDA_RESET/Core_OT
Bit 13	parallel_mode_sel_09_10 0=disable 1=enable	0	POR/RESETn/SPI Time out Latch/ WDA_RESET/Core_OT
Bit 12	parallel_mode_sel_07_08 0=disable 1=enable	0	POR/RESETn/SPI Time out Latch/ WDA_RESET/Core_OT
Bit 11	parallel_mode_sel_05_06 0=disable 1=enable	0	POR/RESETn/SPI Time out Latch/ WDA_RESET/Core_OT
Bit 10	parallel_mode_sel_04_03 0=disable 1=enable	0	POR/RESETn/SPI Time out Latch/ WDA_RESET/Core_OT
Bit 9	parallel_mode_sel_02_01 0=disable 1=enable	0	POR/RESETn/SPI Time out Latch/ WDA_RESET/Core_OT
Bit 8:7	inx_fit_config<1:0> 00:0 μ s 01:0.5 μ s 10:1 μ s 11:2 μ s	0	POR/RESETn/SPI Time out Latch/ WDA_RESET/Core_OT
Bit 6	Spread spectrum disable 0=disable 1=enable	0	POR/RESETn/SPI Time out Latch/ WDA_RESET/Core_OT
Bit 5	self_test_en 0=disable 1=enable	0	POR/RESETn/SPI Time out Latch/ WDA_RESET/Core_OT
Bit 4	csa_enable 0=disable 1=enable	1	POR/RESETn/SPI Time out Latch/ WDA_RESET/Core_OT
Bit 3	frm_cnt_dis 0=enable 1= disable	0	POR/RESETn/SPI Time out Latch/ WDA_RESET/Core_OT
Bit 2:0	Not used	0	POR/RESETn/SPI Time out Latch/ WDA_RESET/Core_OT

Servflt mask:

ADDRESS	Name	TYPE	Notes
00000110	Servflt mask	R/W	

Table 10-12 servflt mask

Data field	Description	Reset value	Reset event
Bit 15:12	Not used	0	POR/RESETn/SPI Time out Latch
Bit 11	wda_reset_mask	0	POR/RESETn/SPI

	0=not masked 1=masked		Time out Latch
Bit 10	en_dr_echo_mask 0=not masked 1=masked	0	POR/RESETn/SPI Time out Latch
Bit 9	spi_timeout_latch_mask 0=not masked 1=masked	0	POR/RESETn/SPI Time out Latch
Bit 8	spi_timeout_mask 0=not masked 1=masked	0	POR/RESETn/SPI Time out Latch
Bit 7	spi_protocol_err_mask 0=not masked 1=masked	0	POR/RESETn/SPI Time out Latch
Bit 6	vdd_uv_mask 0=not masked 1=masked	0	POR/RESETn/SPI Time out Latch
Bit 5	Not used	0	POR/RESETn/SPI Time out Latch
Bit 4	cp_uv_mask 0=not masked 1=masked	0	POR/RESETn/SPI Time out Latch
Bit 3	NOT USED	0	POR/RESETn/SPI Time out Latch
Bit 2	vbat_uv_mask 0=not masked 1=masked	0	POR/RESETn/SPI Time out Latch
Bit 1	vbat_ov_mask 0=not masked 1=masked	0	POR/RESETn/SPI Time out Latch
Bit 0	core_ot_mask 0=not masked 1=masked	0	POR/RESETn/SPI Time out Latch

Dervfltsk:

ADDRESS	Name	TYPE	Notes
00000111	dervfltsk	R/W	

Table 10-13 dervfltsk

Data field	Description	Reset value	Reset event
Bit 15:9	NOT USED	0	POR/RESETn/SPI Time out Latch
Bit 8	sol_gndloss_mask 0=not masked 1=masked	0	POR/RESETn/SPI Time out Latch
Bit 7	sol_otw_mask 0=not masked 1=masked	0	POR/RESETn/SPI Time out Latch
Bit 6	sol_pwm_check_mask 0=not masked 1=masked	0	POR/RESETn/SPI Time out Latch
Bit 5	sol_opl_fault_mask 0=not masked 1=masked	0	POR/RESETn/SPI Time out Latch

Bit 4	sol_short_fault_mask 0=not masked 1=masked	0	POR/RESETn/SPI Time out Latch
Bit 3	sol_lscamp_fault_mask 0=not masked 1=masked	0	POR/RESETn/SPI Time out Latch
Bit 2	sol_hs_oc_fault_mask 0=not masked 1=masked	0	POR/RESETn/SPI Time out Latch
Bit 1	sol_ls_oc_fault_mask 0=not masked 1=masked	0	POR/RESETn/SPI Time out Latch
Bit 0	sol_ot_fault_mask 0=not masked 1=masked	0	POR/RESETn/SPI Time out Latch

Timeout spi cfg:

ADDRESS	Name	TYPE	Notes
00001000	Timeout spi cfg	R/W	

Table 10-14 timeout spi cfg

Data field	Description	Reset value	Reset event
Bit 15:7	Not used	0	POR/RESETn/SPI Time out Latch
Bit 6:5	timeout_NR RESET<1:0> 00: 2 01: 4 10: 8 11: 16	01	POR/RESETn/SPI Time out Latch
Bit 4:2	timeout_configuration<2:0> 000: 504 μ s 001: 1008 μ s 010: 2016 μ s 011: 4032 μ s 100: 8064 μ s 101: 16128 μ s 110: 32256 μ s 111: 64512 μ s	010	POR/RESETn/SPI Time out Latch
Bit 1	enable reset latch	0	POR/RESETn/SPI Time out Latch
Bit 0	disable reset latch	1	POR/RESETn/SPI Time out Latch

solenoid pwm configuration 0-13:

ADDRESS	Name	TYPE	Notes
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00100000-00101101	solenoid pwm configuration 0	R/W	
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Table 10-15 solenoid pwm configuration 0

Data field	Description	Reset value	Reset event
Bit 15:8	pwm_duty_config_00<7:0> 0000 0000 : 0% ... 0110 0100 : 50% ... 1100 1000 : 100% ... 1111 1111 : 100%	0	POR/RESETn/SPI Time out Latch/ WDA_RESET/Core_OT
Bit 7:4	pwm_freq_sel_00<3:0> 0000 : 1.0k 0001 : 1.5k 0010 : 2.0k 0011 : 2.5k 0100 : 3.5k 0101 : 4.0k 0110 : 4.5k 0111 : 8.0k 1000 : 9.5k 1001 : 10.0k 1010 : 10.5k 1011 : 15.50k 1100 : 16.0k 1101 : 16.5k 1110 : 18.0k 1111 : 20.0k	0	POR/RESETn/SPI Time out Latch/ WDA_RESET/Core_OT
Bit 3:0	not used	0	POR/RESETn/SPI Time out Latch/ WDA_RESET/Core_OT

solenoid driver configuration 0-13:

ADDRESS	Name	TYPE	Notes
00101110-00111011	solenoid driver configuration	R/W	

Table 10-16 solenoid driver configuration

Data field	Description	Reset value	Reset event
Bit 15	drv_hs_ls_sel_00 (Only solenoid 0-3) 0: LS	0	POR/RESETn/SPI Time out Latch/ WDA_RESET/Core_OT

Bit 14:13	1: HS Not used	0	POR/RESETn/SPI Time out Latch/ WDA_RESET/Core_OT
Bit 12	off_diagnostic_enable_00 0=disable 1=enable	0	POR/RESETn/SPI Time out Latch/ WDA_RESET/Core_OT
Bit 11:10	drv_blank_time_sel_00<1:0> 00=384 μ s 01=768 μ s 10=6.25ms 11=3.125ms	1	POR/RESETn/SPI Time out Latch/ WDA_RESET/Core_OT
Bit 9	drv_oc_threshold_sel_00 0=3A 1=4.3A	0	POR/RESETn/SPI Time out Latch/ WDA_RESET/Core_OT
Bit 8:7	drv_sr_config_00<1:0> 00=1v/ μ s 01=4v/ μ s 10=10v/ μ s 11=20v/ μ s	0	POR/RESETn/SPI Time out Latch/ WDA_RESET/Core_OT
Bit 6	PWM_Enable 0=disable 1=enable	0	POR/RESETn/SPI Time out Latch/ WDA_RESET/Core_OT
Bit 5:4	FM_LSD_MP[1:0] 00 = 8 cycle 01 = 16 cycle 10 = 32 cycle 11 = 64 cycle	10	POR/RESETn/SPI Time out Latch/ WDA_RESET/Core_OT
Bit 3:0	FM_LSD_MB[3:0] 0000 = no modulation (default) 0001 = 0.5%/1 step 0010 = 1%/2 step ... 1111 = 7.5%/15 step	0	POR/RESETn/SPI Time out Latch/ WDA_RESET/Core_OT

IC_VERSION:

ADDRESS	Name	TYPE	Notes
01000000	IC VERSION	R	

Table 10-17 IC_VERSION

Data field	Description	Reset value	Reset event
Bit 15:6	not used	0	POR/RESETn/SPI Time out Latch
Bit 5:3	Chip id version<2:0>	010	POR/RESETn/SPI Time out Latch
Bit 2:0	chip id subversion <2:0>	001	POR/RESETn/SPI Time out Latch

10.8 Fault and test page summary

pin status:

ADDRESS	Name	TYPE	Notes
01000001	Input pin status	R	

Table 10-18 pin status

Data field	Description	Reset value	Reset event
Bit15:13	Not used	0	POR/RESETn/SPI Time out Latch
Bit12	fault echo FAUL Pin echo - FAUL pin status feedback	0	POR/RESETn/SPI Time out Latch
Bit11	res_echo 1: RES pin is Low 0: RES pin is high	0	POR/RESETn/SPI Time out Latch
Bit10	in10_echo 0: Logic low 1: Logic high	0	POR/RESETn/SPI Time out Latch
Bit9	In9_echo 0: Logic low 1: Logic high	0	POR/RESETn/SPI Time out Latch
Bit8	In8_echo 0: Logic low 1: Logic high	0	POR/RESETn/SPI Time out Latch
Bit7	In7_echo 0: Logic low 1: Logic high	0	POR/RESETn/SPI Time out Latch
Bit6	In6_echo 0: Logic low 1: Logic high	0	POR/RESETn/SPI Time out Latch
Bit5	In5_echo 0: Logic low 1: Logic high	0	POR/RESETn/SPI Time out Latch
Bit4	In4_echo 0: Logic low 1: Logic high	0	POR/RESETn/SPI Time out Latch
Bit3	In3_echo 0: Logic low 1: Logic high	0	POR/RESETn/SPI Time out Latch
Bit2	In2_echo 0: Logic low 1: Logic high	0	POR/RESETn/SPI Time out Latch

Bit1	In1_echo 0: Logic low 1: Logic high	0	POR/RESETn/SPI Time out Latch
Bit0	en_dr_echo 0: Logic low 1: Logic high	0	POR/RESETn/SPI Time out Latch

odtests 0:

ADDRESS	Name	TYPE	Notes
01000010	auto self-test result	R	

Table 10-19 odtests 0

Data field	Description	Reset value	Reset event
Bit 15:10	not used	0	
Bit 9	vbat_ov2_self_test 0=pass 1=fail	0	POR/RESETn/SPI Time out Latch
Bit 8	vbat_ov_self_test 0=pass 1=fail	0	POR/RESETn/SPI Time out Latch
Bit7	vbat_uv_self_test 0=pass 1=fail	0	POR/RESETn/SPI Time out Latch
Bit6	cp_uv_self_test 0=pass 1=fail	0	POR/RESETn/SPI Time out Latch
Bit5	vdd_ov_self_tests 0=pass 1=fail	0	POR/RESETn/SPI Time out Latch
Bit4	vdd_uv_self_tests 0=pass 1=fail	0	POR/RESETn/SPI Time out Latch
Bit3	v3v3d_ov_self_tests 0=pass 1=fail	0	POR/RESETn/SPI Time out Latch
Bit2	v3v3d_uv_self_tests 0=pass 1=fail	0	POR/RESETn/SPI Time out Latch
Bit1	v3v3a_ov_self_tests 0=pass 1=fail	0	POR/RESETn/SPI Time out Latch
Bit0	v3v3a_uv_self_tests 0=pass 1=fail	0	POR/RESETn/SPI Time out Latch

odtests 1:

ADDRESS	Name	TYPE	Notes
01000011	self-test result	R	

Table 10-20 odttests 1

Data field	Description	Reset value	Reset event
Bit 15:12	not used	0	
Bit11	gndloss_self_test_11 0=pass 1=fail	0	POR/RESETn/SPI Time out Latch
Bit10	gndloss_self_test_10 0=pass 1=fail	0	POR/RESETn/SPI Time out Latch
Bit 9	gndloss_self_test_09 0=pass 1=fail	0	POR/RESETn/SPI Time out Latch
Bit 8	gndloss_self_test_08 0=pass 1=fail	0	POR/RESETn/SPI Time out Latch
Bit7	gndloss_self_test_07 0=pass 1=fail	0	POR/RESETn/SPI Time out Latch
Bit6	gndloss_self_test_06 0=pass 1=fail	0	POR/RESETn/SPI Time out Latch
Bit5	gndloss_self_test_05 0=pass 1=fail	0	POR/RESETn/SPI Time out Latch
Bit4	gndloss_self_test_04 0=pass 1=fail	0	POR/RESETn/SPI Time out Latch
Bit3	gndloss_self_test_03 0=pass 1=fail	0	POR/RESETn/SPI Time out Latch
Bit2	gndloss_self_test_02 0=pass 1=fail	0	POR/RESETn/SPI Time out Latch
Bit1	gndloss_self_test_01 0=pass 1=fail	0	POR/RESETn/SPI Time out Latch
Bit0	gndloss_self_test_00 0=pass 1=fail	0	POR/RESETn/SPI Time out Latch

odtest2:

ADDRESS	Name	TYPE	Notes
01000100	self-test result	R	

Table 10-21 odttests 2

Data field	Description	Reset value	Reset event
Bit 15:14	drv_diag_self_test_07<1:0> 00: Self Test pass 01: DIAG_LV comparator fail 10: DIAG_OL comparator fail	0	POR/RESETn/SPI Time out Latch

	11: Self Test fail		
Bit 13:12	drv_diag_self_test_06<1:0> 00: Self Test pass 01: DIAG_LV comparator fail 10: DIAG OL comparator fail 11: Self Test fail	0	POR/RESETn/SPI Time out Latch
Bit 11:10	drv_diag_self_test_05<1:0> 00: Self Test pass 01: DIAG_LV comparator fail 10: DIAG OL comparator fail 11: Self Test fail	0	POR/RESETn/SPI Time out Latch
Bit 9:8	drv_diag_self_test_04<1:0> 00: Self Test pass 01: DIAG_LV comparator fail 10: DIAG OL comparator fail 11: Self Test fail	0	POR/RESETn/SPI Time out Latch
Bit 7:6	drv_diag_self_test_03<1:0> 00: Self Test pass 01: DIAG_LV comparator fail 10: DIAG OL comparator fail 11: Self Test fail	0	POR/RESETn/SPI Time out Latch
Bit 5:4	drv_diag_self_test_02<1:0> 00: Self Test pass 01: DIAG_LV comparator fail 10: DIAG OL comparator fail 11: Self Test fail	0	POR/RESETn/SPI Time out Latch
Bit 3:2	drv_diag_self_test_01<1:0> 00: Self Test pass 01: DIAG_LV comparator fail 10: DIAG OL comparator fail 11: Self Test fail	0	POR/RESETn/SPI Time out Latch
Bit 1:0	drv_diag_self_test_00<1:0> 00: Self Test pass 01: DIAG_LV comparator fail 10: DIAG OL comparator fail 11: Self Test fail	0	POR/RESETn/SPI Time out Latch

odtest3:

ADDRESS	Name	TYPE	Notes
01000101	self-test result	R/W	POR/RESETn/SPI Time out Latch

Table 10-22 odttests 3

Data field	Description	Reset value	Reset event
Bit 15:12	not used	0	POR/RESETn/SPI Time out Latch
Bit 11:10	drv_diag_self_test_13<1:0> 00: Self Test pass 01: DIAG_LV comparator fail 10: DIAG OL comparator fail 11: Self Test fail	0	POR/RESETn/SPI Time out Latch
Bit 9:8	drv_diag_self_test_12<1:0> 00: Self Test pass 01: DIAG_LV comparator fail 10: DIAG OL comparator fail 11: Self Test fail	0	POR/RESETn/SPI Time out Latch
Bit 7:6	drv_diag_self_test_11<1:0> 00: Self Test pass 01: DIAG_LV comparator fail 10: DIAG OL comparator fail 11: Self Test fail	0	POR/RESETn/SPI Time out Latch
Bit 5:4	drv_diag_self_test_10<1:0> 00: Self Test pass 01: DIAG_LV comparator fail 10: DIAG OL comparator fail 11: Self Test fail	0	POR/RESETn/SPI Time out Latch
Bit 3:2	drv_diag_self_test_09<1:0> 00: Self Test pass 01: DIAG_LV comparator fail 10: DIAG OL comparator fail 11: Self Test fail	0	POR/RESETn/SPI Time out Latch
Bit 1:0	drv_diag_self_test_08<1:0> 00: Self Test pass 01: DIAG_LV comparator fail 10: DIAG OL comparator fail 11: Self Test fail	0	POR/RESETn/SPI Time out Latch

servflt1:

ADDRESS	Name	TYPE	Notes
01000110	system diagnostic	R	

Table 10-23 servflt1

Data field	Description	Reset value	Reset event
Bit15	Not used	0	POR/RESETn/SPI Time out Latch
Bit14	vbat_over_voltage2 0=pass 1=fail	0	POR/RESETn/SPI Time out Latch
Bit13	vbat_over_voltage2_det (not latch) 0=pass 1=fail	0	Right vbat access
Bit12	efuse_check_ok 1=pass 0=fail	0	POR/RESETn/SPI Time out Latch
Bit11	por_flag 0=pass 1=fail	0	POR/RESETn/SPI Time out Latch
Bit10	diag_osc_sa_monitor 0=pass 1=fail	0	POR/RESETn/SPI Time out Latch
Bit9	main_osc_monitor 0=pass 1=fail	0	POR/RESETn/SPI Time out Latch
Bit8	spi_timeout_latch 0=pass 1=fail	0	POR/RESETn/SPI Time out Latch
Bit7	spi_timeouth 0=pass 1=fail	0	POR/RESETn/SPI Time out Latch
Bit6	vdd_over_voltage 0=pass 1=fail	0	POR/RESETn/SPI Time out Latch
Bit5	vdd_under_voltage 0=pass 1=fail	0	POR/RESETn/SPI Time out Latch
Bit4	cp_under_voltage 0=pass 1=fail	0	POR/RESETn/SPI Time out Latch
Bit3	Not used	0	POR/RESETn/SPI Time out Latch
Bit2	vbat_under_voltage 0=pass 1=fail	0	POR/RESETn/SPI Time out Latch

Bit1	vbat_over_voltage 0=pass 1=fail	0	POR/RESETn/SPI Time out Latch
Bit0	core_over_temperature 0=pass 1=fail	0	POR/RESETn/SPI Time out Latch

Communication flts:

ADDRESS	Name	TYPE	Notes
01000111	spi fault & v3v3 fault	R	

Table 10-24 Communication flts

Data field	Description	Reset value	Reset event
Bit15:13	Not used	0	POR/RESETn/SPI Time out Latch
Bit12	frame_counter_error_flag 0=pass 1=fail	0	Right SPI access
Bit11	crc error flag 0=pass 1=fail	0	Right SPI access
Bit10	short_frame_err 0=pass 1=fail	0	Right SPI access
Bit9	long_frame_err_flag 0=pass 1=fail	0	Right SPI access
Bit8:4	Not used	0	POR/RESETn/SPI Time out Latch
Bit3	v3v3d_under_voltage 0=pass 1=fail	0	POR/RESETn/SPI Time out Latch
Bit2	v3v3d_over_voltage 0=pass 1=fail	0	POR/RESETn/SPI Time out Latch
Bit1	v3v3a_under_voltage 0=pass 1=fail	0	POR/RESETn/SPI Time out Latch
Bit0	v3v3a_over_voltage 0=pass 1=fail	0	POR/RESETn/SPI Time out Latch

Temperature diagnostic:

ADDRESS	Name	TYPE	Notes
01001000	Temperature warning	R	

Table 10-25 Temperature diagnostic

Data field	Description	Reset value	Reset event
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Bit15:14	Not used	0	POR/RESETn/SPI Time out Latch
Bit13:0	T_warning<13:0>	0	POR/RESETn/SPI Time out Latch

Temperature diagnostic:

ADDRESS	Name	TYPE	Notes
01001001	Temperature shutdown	R	POR/RESETn/SPI Time out Latch

Table 10-26 Temperature diagnostic

Data field	Description	Reset value	Reset event
Bit15:14	Not used	0	POR/RESETn/SPI Time out Latch
Bit13:0	T_SD<13:0>	0	POR/RESETn/SPI Time out Latch

Note: Temperature diagnosis of channels 1-14 mapped by T_warning<13:0> and T_SD<13:0>.

10.9 Channel page registers**exceptions0-3:**

ADDRESS	Name	TYPE	Notes
01100000-01100011	sol driver channel 1-4 diagnostic result	R	

Table 10-27 exceptions 0-3

Data field	Description	Reset value	Reset event
Bit15:9	Not used	0	POR/RESETn/SPI Time out Latch
Bit8	gndloss_fault	0	POR/RESETn/SPI Time out Latch
Bit7	pwm_check_comp_mismatch 0=pass 1=fail	0	POR/RESETn/SPI Time out Latch
Bit6	pwm_check_fault 0=pass 1=fail	0	POR/RESETn/SPI Time out Latch
Bit5	Open_load_ 0=pass 1=fail	0	POR/RESETn/SPI Time out Latch
Bit4	short_detection 0=pass 1=fail	0	POR/RESETn/SPI Time out Latch

Bit3	ls_clamp_active 0=pass 1=fail	0	POR/RESETn/SPI Time out Latch
Bit2	hs_over_current 0=pass 1=fail	0	POR/RESETn/SPI Time out Latch
Bit1	ls_over_current 0=pass 1=fail	0	POR/RESETn/SPI Time out Latch
Bit0	Not used	0	POR/RESETn/SPI Time out Latch

Exceptions 4-13:

ADDRESS	Name	TYPE	Notes
01100100-01101101	sol driver channel 5-14 diagnostic result	R	

Table 10-28 exceptions4-13

Data field	Description	Reset value	Reset event
Bit15:9	Not used	0	POR/RESETn/SPI Time out Latch
Bit8	gndloss_fault	0	POR/RESETn/SPI Time out Latch
Bit7	pwm_check_comp_mismatch 0=pass 1=fail	0	POR/RESETn/SPI Time out Latch
Bit6	pwm_check_fault 0=pass 1=fail	0	POR/RESETn/SPI Time out Latch
Bit5	openload_ 0=pass 1=fail	0	POR/RESETn/SPI Time out Latch
Bit4	short_detection 0=pass 1=fail	0	POR/RESETn/SPI Time out Latch
Bit3	ls_clamp_active 0=pass 1=fail	0	POR/RESETn/SPI Time out Latch
Bit2	Not used	0	POR/RESETn/SPI Time out Latch
Bit1	ls_over_current 0=pass 1=fail	0	POR/RESETn/SPI Time out Latch
Bit0	Not used	0	POR/RESETn/SPI Time out Latch

10.10 Q&A watchdog**WD_SEED:**

ADDRESS	Name	TYPE	Notes
10000000	WD_SEED	R	

Table 10-29 WD_SEED

Data field	Description	Reset value	Reset event
Bit15:8	Not used	0	POR/RESETn/SPI Time out Latch
Bit7:0	SEED	0	POR/RESETn/SPI Time out Latch

WD_ANSW:

ADDRESS	Name	TYPE	Notes
10000001	WD_ANSW	R/W	

Table 10-30 WD_ANSW

Data field	Description	Reset value	Reset event
Bit15:8	ANSW HIGH	0	POR/RESETn/SPI Time out Latch
Bit7:0	ANSW LOW	0	POR/RESETn/SPI Time out Latch

WD_STATUS_REGISTER:

ADDRESS	Name	TYPE	Notes
10000010	WD_STATUS_REGISTER	R	

Table 10-31 WD_STATUS_REGISTER

Data field	Description	Reset value	Reset event
Bit15:12	WD_RST_EVENT_VALUE[3:0]	0	POR/RESETn/SPI Time out Latch
Bit11	WD_RST_CNT 0: No fault 1: Fault	0	POR/RESETn/SPI Time out Latch
Bit10	WD_RST_TO_REQ 0: No fault 1: Fault	0	POR/RESETn/SPI Time out Latch
Bit9	WD_RST_TO_ANSW 0: No fault 1: Fault	0	POR/RESETn/SPI Time out Latch
Bit8	WD_LATE_REQ 0: No fault 1: Fault	0	POR/RESETn/SPI Time out Latch

Bit7	WD_EARLY_REQ 0: No fault 1: Fault	0	POR/RESETn/SPI Time out Latch
Bit6	WD_BAD_ANSW 0: No fault 1: Fault	0	POR/RESETn/SPI Time out Latch
Bit5	WD_LATE_ANSW 0: No fault 1: Fault	0	POR/RESETn/SPI Time out Latch
Bit4	WD_EARLY_ANSW 0: No fault 1: Fault	0	POR/RESETn/SPI Time out Latch
Bit3:0	WD_CNT_VALUE	0111	POR/RESETn/SPI Time out Latch

WD_ANSW_TIMING:

ADDRESS	Name	TYPE	Notes
1000011	WD_ANSW_TIMING	R/W	WD_reset not clear WD information

Table 10-32 WD_ANSW_TIMING

Data field	Description	Reset value	Reset event
Bit 15	Not used	0	POR/RESETn/SPI Time out Latch
Bit 14	WD_rst_en	1	POR/RESETn/SPI Time out Latch
Bit 13:8	T_VALID_ANSW_END_DELTA	110000	POR/RESETn/SPI Time out Latch
Bit 7:0	T_VALID_ANSW_START	11111111	POR/RESETn/SPI Time out Latch

WD_REQ_TIMING:

ADDRESS	Name	TYPE	Notes
10000100	WD_REQ_TIMING	R/W	

Table 10-33 WD_REQ_TIMING

Data field	Description	Reset value	Reset event
Bit 15	not used	0	POR/RESETn/SPI Time out Latch
Bit 14	WD_REQ_CHECK_EN	0	POR/RESETn/SPI Time out Latch
Bit 13:8	T_VALID_REQ_END_DELTA	110000	POR/RESETn/SPI Time out Latch
Bit 7:0	T_VALID_REQ_START	11111111	POR/RESETn/SPI Time out Latch

WD_COUNTER_SETUP:

ADDRESS	Name	TYPE	Notes
10000101	WD_COUNTER_SETUP	R/W	

Table 10-34 WD_COUNTER_SETUP

Data field	Description	Reset value	Reset event
Bit 15	WD_TO_RST_EN	0	POR/RESETn/SPI Time out Latch
Bit 14	WD_CLK_DIV	0	POR/RESETn/SPI Time out Latch
Bit 13:10	WD_TH_LOW	0111	POR/RESETn/SPI Time out Latch
Bit 9:6	WD_TH_HIGH	1111	POR/RESETn/SPI Time out Latch
Bit 5:3	WD_CNT_GOOD_STEP	001	POR/RESETn/SPI Time out Latch
Bit 2:0	WD_CNT_BAD_STEP	011	POR/RESETn/SPI Time out Latch

WD_CONFIG:

ADDRESS	Name	TYPE	Notes
10000110	WD_CONFIG	R/W	

Table 10-35 WD_CONFIG

Data field	Description	Reset value	Reset event
Bit 15: 10	T_VALID_ANSW_TO_DELTA	0	POR/RESETn/SPI Time out Latch
Bit 9:4	T_VALID_REQ_TO_DELTA	0	POR/RESETn/SPI Time out Latch
Bit 3	not used	0	POR/RESETn/SPI Time out Latch
Bit 2	wd_enable 0=disable 1=enable	1	POR/RESETn/SPI Time out Latch
Bit 1	disable uc reset (inhibit fault _low from WDA) 0=disable 1=enable	0	POR/RESETn/SPI Time out Latch
Bit 0	disable driver inhibit 0=disable 1=enable	0	POR/RESETn/SPI Time out Latch

HE9314

Automotive 14 channels valve driver



Recommended pad size

11 Package

11.1 ETQFP package information

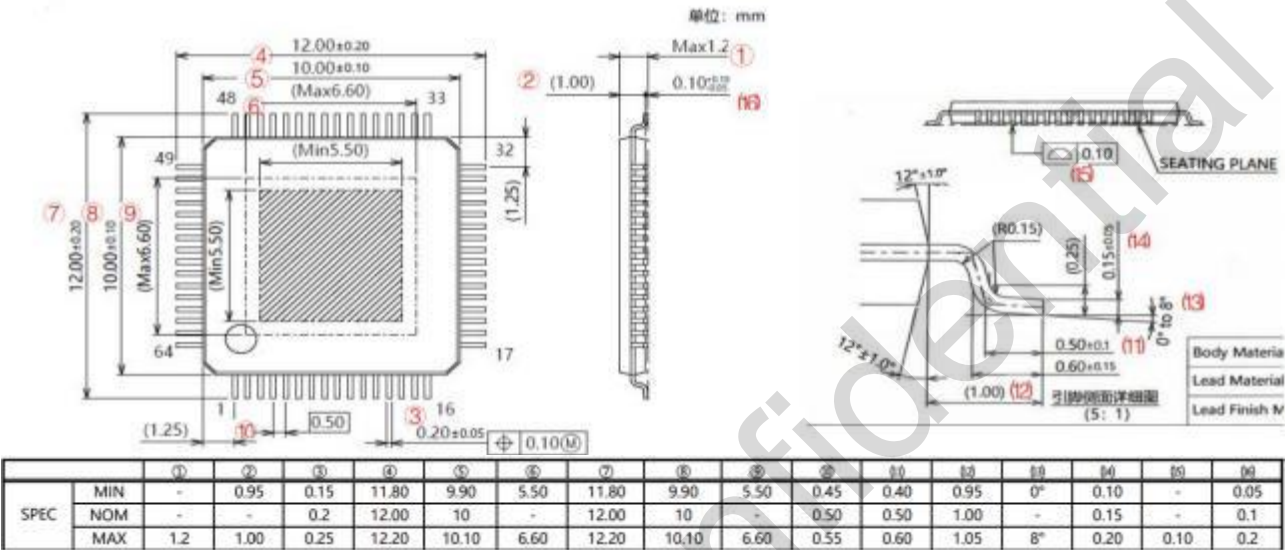
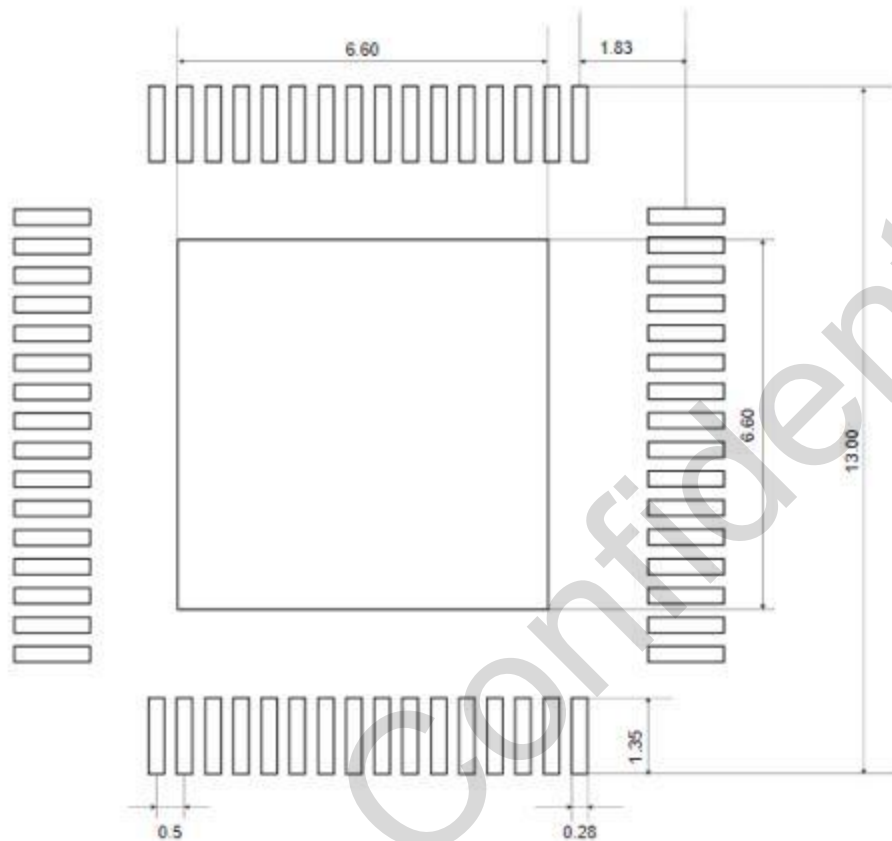


Figure 11-1 ETQFP package dimension

12 Recommended pad size



Unit: mm

Figure 12-1 Recommended pad size

Notes: information of Recommended pad size is for reference only and can be adjusted according to needs.

13 Customer Service

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14 Revision history

Date	Revision	Changes
2026/3/05	1.0	1、optimized layout. 2、Update Part Number. 3、Deleted the bit for configuring the closed-loop current 4、optimize parameters.